

TrackMaster Pro

User Manual · Version 1.5.0

Track planning for model railway layouts in all common gauges – with height planning, helix and timber list, modules, levels and benchwork, scenery, room planning, wiring and signals, plan check, costs and 3D view. For Windows, Android and the browser – with a step-by-step introduction for everyone planning for the first time.

TrackMaster Pro 1.5.0 · Basic and Pro

As of November 2026

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1 Welcome

TrackMaster Pro is a planning program for model railway layouts. You draw your layout on screen before you buy the first rail or saw the first batten: baseboard, track, gradients, scenery, power and signals. The program works out the dimensions for you, warns you about mistakes and shows the result in 3D.

What TrackMaster Pro can do

You do not have to use everything at once. For a first small layout, the baseboard, the track and the plan check are enough. Everything else can come later, when you need it.

Area	What you do with it
Track	Track plans in gauges Z, N, TT, H0, H0m, H0e, 0, gauge 1 and G. Numerous track systems with around 900 pieces are available: straights, curves, turnouts, crossings, double slips, buffer stops, turntables and flexible track.
Baseboard	Rectangle, L, U, T, Z, E and G shape or a ring with an open centre. Every leg can be sized individually. A layout can also consist of **modules**.
Benchwork	Wooden frame with legs, with material and cutting list; open-frame construction with ribs and roadbeds.
Heights	Gradients over whole runs of track, up to 25 levels on top of each other, round or oval helixes, single or double track, with timber list.
Scenery	Hills, mountains, lakes, rivers, buildings, trees and tunnels.
Scenery items	20 ready-made models: tunnel portals, platforms, bridges, station, engine shed, houses. They snap to the track. With Pro you build your own models in the model editor.
Room	Your room with walls, doors, windows, radiators and furniture around the layout. A scanned floor plan or track plan can lie under the plan as a background image.
Wiring	Operation (digital or analog), command station, booster and transformer, wires with feeders and ground, isolating joints, reverse loops, signals from several countries, feedback sensors, digital addresses, wiring plan and routes.
Checking	The plan check reports tracks that are too close, ramps that are too steep, track over the baseboard edge, open track ends, curves too tight for long coaches, aisles too narrow and more. The wiring check reports sections without power, missing ground and short circuits.
3D view	The layout from every side, even from below. You can ride along in the cab and set the time of day.

Results PRO	Parts list with prices and your own stock, printing (also as a full-size building template), export as image, vector graphic and CAD drawing, helix timber list, rib construction and roadbeds.

Basic and Pro

TrackMaster Pro comes in two editions. Without a licence the **Basic edition** runs. It contains **all planning functions**: you can design, check and view any layout in 3D. The Basic edition saves plans with up to **50 track pieces**. A licence turns it into the **Pro edition**. It saves plans of any size and unlocks the results of your planning: parts list, printing and exports.

Basic (no licence)	Pro (with licence) – in addition
Track planning in all gauges with all track systems, flexible track	Saving plans of any size (Basic: up to 50 tracks)
Baseboard shapes, modules, leg dimensions, levels, benchwork with material list	Parts list of track and accessories, also as a CSV file
Height planning: gradient, round and oval helix, single and double track	Rib construction and roadbed from selection, with SVG cutting templates
Terrain, water, buildings, vegetation, tunnels	Helix timber list with all cutting templates, also as SVG
Wiring: command station, wires, feeders, isolating joints, signals, addresses, wiring plan	Printing the track plan, also as a 1:1 building template
3D view with ride-along and time of day, plan and wiring check	Export as SVG at 1:1, wiring plan as a text file
Room with doors, windows and furniture, background image	Parts list with prices, stock, “still to buy” and price list
Parallel track, height profile, lock, group, align, feedback sensors	Export as PNG image and as DXF drawing for CAD programs
Scenery items: 20 basic models, models saved with the plan	Model editor, own model library, import, export as STL and OBJ
Opening plans, viewing routes	Export for control software (TXT, CSV, JSON)

Pro functions are marked in the program: in the tooltip as “Pro function”, in the Android app with a padlock. If you choose a Pro function in the Basic edition, a note explains what Pro offers and, if you wish, takes you straight to the licence. Your plan stays open and unchanged.

Three versions: Windows, browser and Android

TrackMaster Pro runs as a program on **Windows**, as a **web version** in any current browser and as an **app on Android**. All three open the same plans and have the same tabs. A few things differ:

What	Windows	Web version (browser)

File commands	Green **File** button at the top left opens a menu	**File** tab with the groups Plan, Output and Program
Undo and history	**Home › History** and the quick access bar at the very top	Arrows and clock icon at the top right of the title bar
Checking the plan	**Home › Collision check**	**Home › Plan check**
Loading the sample	**File › Load sample plan**	**File › Sample layout**
Snapping on/off	**Snap** switch in the status bar	**Snap** switch in the status bar or under **View**
Help on keys	**View › Shortcuts** (F1)	**View › How to use** or **File › How to use** (F1)
Plan file	Extension **.trackmaster.json**	Extension **.mpr** (same content; also opens .trackmaster.json)
Only here	–	Layout suggestion from your own track stock, turntable as a quick command, modules at different heights, cutting track at module edges, changing the height at a joint

The Android app is designed for finger operation. What is different there is described in chapter 17.

How to read this manual

> **New to this?**
> Then start with chapter 4 “First steps for beginners”. It explains the basic terms and how to use the mouse and keyboard, and takes you to your first layout in ten steps. Read the other chapters later, one at a time, when you need a function.

- **Commands** on the ribbon are written like this: **Tab › Command**. **Tracks › Gradient…** means: click the **Tracks** tab at the top, then the **Gradient…** button.
- **Keys** look like this: Ctrl+S. That means: hold down Ctrl and then press S.
- **Three dots** after a command (for example **Room…**) mean that a window opens first in which you enter something.
- **Dimensions** are given in **millimetres** throughout the program. 1 m is 1000 mm, 1 cm is 10 mm.
- The pictures show the Windows program. The web version looks almost the same.

> **Tip**
> Boxes like this show shortcuts and proven ways of working.

Caution

Notes marked like this should be observed before building.

Example

In grey boxes we work through something once with real numbers.

2 Installation

You only need one of the three versions. The web version runs without installation; you install the Windows program and the Android app once.

Windows

Requirements

- Windows 10 or Windows 11, 64-bit
- about 400 MB of free disk space
- screen of at least 1280 × 720 pixels, 1920 × 1080 recommended
- for the 3D view, a graphics card with DirectX support (present in every current PC)

You do not need to install anything else. Everything TrackMaster Pro needs is included in the installation package.

Installing

1. Download the file **TrackMasterPro-1.5.0.msi**. It usually ends up in the *Downloads* folder.
2. Double-click the file.
3. If Windows reports “Windows protected your PC”, click **More info** and then **Run anyway**. This message appears for programs that Windows has not seen often yet.
4. The setup wizard welcomes you. Click **Next**.
5. Read the licence terms, tick **I accept the terms** and click **Next**.
6. Keep the suggested destination folder (*C:\Program Files\TrackMaster Pro*) and click **Next** and **Install**. Windows asks whether the program may make changes. Confirm with **Yes**.
7. After a few seconds everything is done. Click **Finish**.

You will then find TrackMaster Pro in the **Start menu** and as a shortcut on the **desktop**. This manual is also in the Start menu.

Updating and uninstalling

Simply install a new version over the old one. The installer detects the older version and replaces it. Your plans and your licence are kept. To remove the program, open **Settings › Apps › Installed apps** in Windows, find **TrackMaster Pro** and click **Uninstall**. Your plan files and the licence file are not deleted.

For administrators: silent installation

Command	Effect
`msiexec /i TrackMasterPro-1.5.0.msi /qn`	Installation without user interface

`msiexec /i TrackMasterPro-1.5.0.msi INSTALLFOLDER="D:\Programs\TrackMaster Pro" /qn`	Installation into another folder
`msiexec /x TrackMasterPro-1.5.0.msi /qn`	Uninstallation without user interface

Web version in the browser

The web version runs in any current browser on a PC or tablet. You simply open the web address. The first time, it loads a few megabytes and starts after a few seconds. After that it comes from the browser's cache and starts faster.

Topic	How it works in the web version
Your plan	The browser keeps remembering the current plan. If you close the window and open the page again, the plan is still there. This does not replace saving a file, though: if you delete the site data, this copy is gone too.
Saving	**File › Save** downloads the plan as a file with the extension **.mpr**, usually into the *Downloads* folder. **File › Save as…** (Ctrl + Shift + S) asks for the file name. In some browsers you also choose the folder, and **Save** then always writes to that file.
Opening	**File › Open…** loads a **.mpr** or **.trackmaster.json** file. The content is the same as on Windows and Android. There you open a .mpr file by choosing “All files” in the open window.
Licence	**File › Licence…**: choose the licence file or paste its content. It applies only in this browser. After deleting the site data, install it again.
3D view	needs WebGL. If the browser reports that it cannot show 3D graphics, turn on hardware acceleration in its settings.
Printing, exports	Files are downloaded. Printing uses the browser's print window, where you can also choose “Save as PDF”.
Own models PRO	are, like the licence, stored only in this browser. Back them up now and then with **Scenery items › Library… › Export all (ZIP)…** (chapter 11).
Home page	If the web version is embedded in a website, the **Home** button at the top left takes you back to that website.

Android

Requirements

- Smartphone or tablet with Android 8.0 or later. A tablet of 10 inches or more is much more comfortable for planning.
- about 100 MB of free storage

Installing

1. Copy the file **TrackMasterPro-1.5.0.apk** to the device, for example via USB cable, cloud storage or as an e-mail attachment.
2. Tap the file on the device, for example in the **Files** app in the **Downloads** folder.
3. The first time, Android asks whether this app may **install unknown apps**. Tap **Settings**, allow it and go back.
4. Tap **Install**. TrackMaster Pro then appears in the app overview.

Install an update the same way over the existing app. Your current plan is kept.

New versions

When it starts, the Windows program briefly asks the download page whether there is a new version and reports it at the bottom in the status bar. Under **File › Check for updates…** you can check yourself at any time and open the download page if you wish. With **File › Update check on start-up…** you switch the check at start-up off or on again. Nothing is ever installed automatically. Without internet, the check is simply skipped. The web version is always up to date as soon as the website has been updated.

3 Licence and activation

The Pro edition is unlocked with a personal licence file. The file is digitally signed and cannot be altered. The program checks it every time it starts.

Installing the licence

1. Save the licence file you received after purchase (extension **.lic**) on the computer, for example in the *Downloads* folder.
2. Open **File › Licence…**. On Windows you can also use **Home › Licence**.
3. Click **Install licence…** and select the file. In the web version you can also paste the file's content into the text box.
4. The window now shows **Pro**, the licensee's name and, if present, the expiry date. The indicator at the top or in the status bar changes from “Basic” to “Pro”.

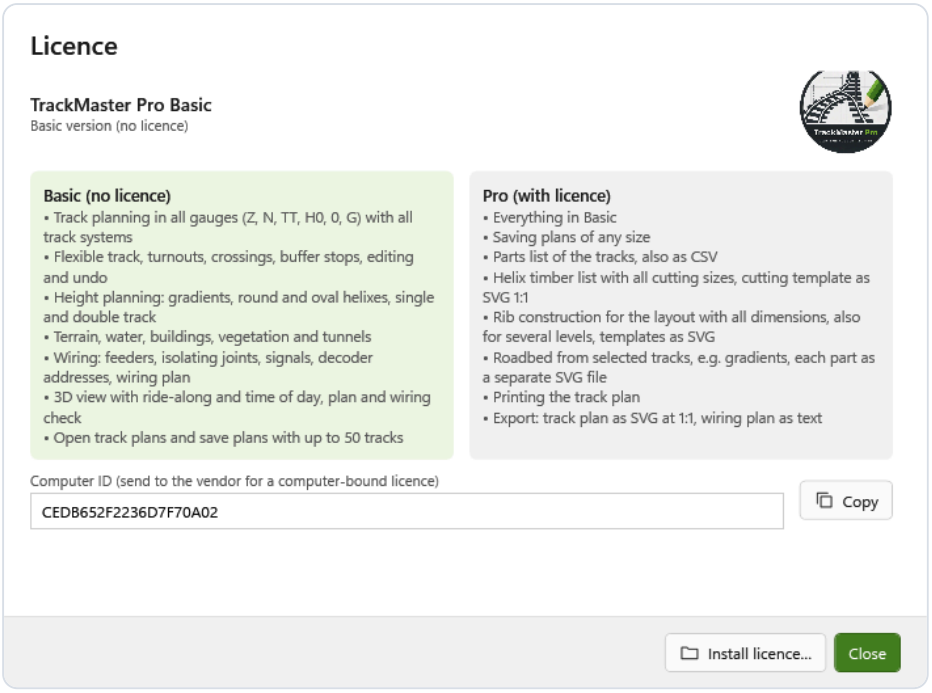


Figure 1 – The licence window shows the edition and what Basic and Pro contain.

On Windows the licence is then stored in your user data (*%APPDATA%\TrackMaster Pro\trackmaster.lic*) and applies at every start. On Android you install the licence the same way via **File › Licence…** and choose the file from the device's storage.

Online activation: up to two devices

A licence without computer binding may run on **two devices** at the same time, for example on the PC at home and on a tablet. Every device counts, whether Windows, Android or browser. For this to work, the device registers once over the internet the first time it starts with the licence. After that it checks about once a week whether the registration still exists. The licence window shows the status in the **Activation** section.

- **Without internet**, Pro keeps running for up to 14 days after installing the licence. A device that has been activated stays activated without internet.
- If **both places are taken**, the third device stays on the Basic edition. The message names the registered devices.
- **Changing devices:** on the old device choose **Release this device** in the licence window. Then click **Activate now** on the new device. If the old device no longer exists (broken, sold), the publisher frees the place on request.
- **Browser:** every browser counts as a separate device. If you delete the site data, it registers as a new device next time.

Licence bound to a computer

A licence can be bound to a particular computer or device. For this the publisher needs the **computer ID** (on Android: **device ID**). It is shown in the licence window. **Copy** puts it on the clipboard; then paste it into your e-mail (Ctrl + V).

If you choose a Pro function in the Basic edition

If you choose the parts list, printing or an export, or save a plan with more than 50 tracks, a note appears. **Licence…** takes you straight to the licence window. Your plan stays open and unchanged.

> **Caution**
> In the Basic edition, plans with more than 50 tracks cannot be saved. If you then close the program, start a new plan or open another one, TrackMaster Pro asks whether the changes should be discarded. Only answer Yes if you really no longer need the plan.

4 First steps for beginners

Have you never planned a model railway on a computer before? Then this chapter is for you. It explains the most important words and how to use the mouse and keyboard, and takes you step by step to your first small layout. You need no previous knowledge and no licence.

The most important terms

A track plan is a drawing of your layout seen from above, to the correct scale. It consists of individual **track pieces**, just as the real layout is made of track you buy. You will meet these words all the time:

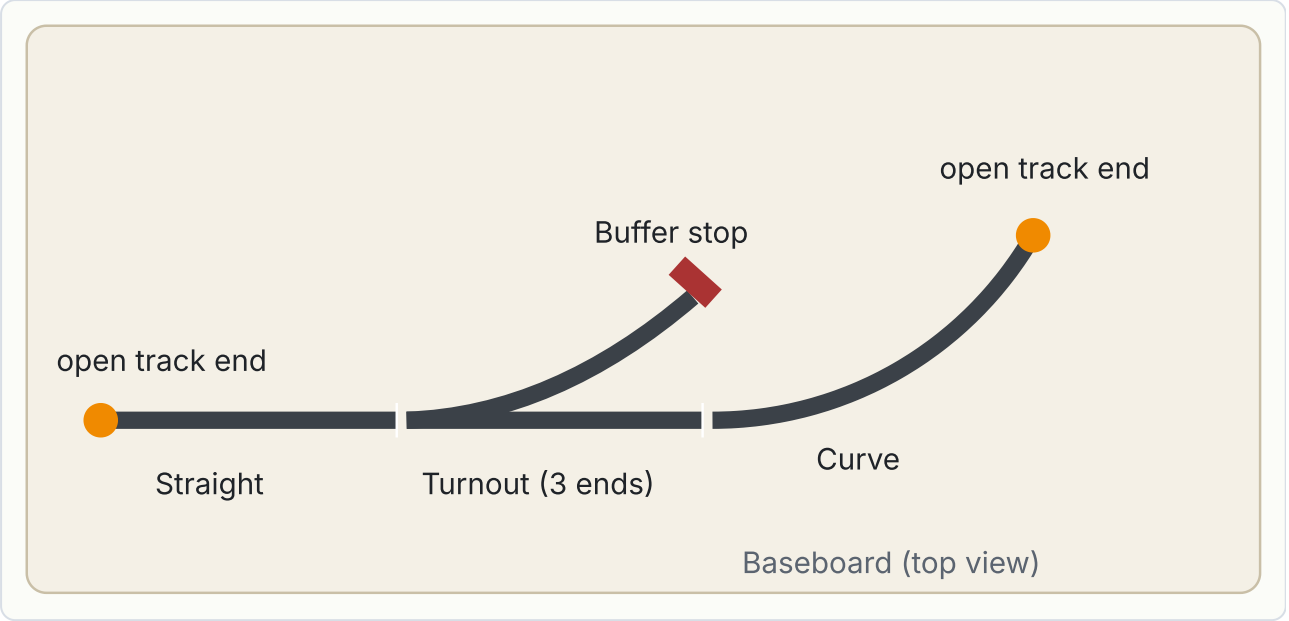


Figure 2 – Straight, turnout, curve and buffer stop; the orange dots are open track ends.

Term	What it means
Gauge	The size of the model railway. The most common is **H0** (scale 1:87). **N** is smaller (1:160), **Z** very small (1:220), **G** large, for the garden. You choose the gauge at the start. It determines which track is available.
Scale	How much smaller the model is than the real thing. At 1:87, 1 cm on the model is 87 cm in reality.
Track system	A manufacturer's track range. Pieces of one system fit together exactly; pieces of different systems usually need an adapter. Choose the system you want to buy or already own.
Track piece	A single part: **straight**, **curve**, **turnout** (where the track divides), **crossing** (two tracks cross) or **buffer stop** (the end of a siding).

Track end	Every track piece has ends where the next one connects. A straight has two, a turnout three. An end without a connection is an **open track end** and is marked **orange** in the plan.
Active track end	The open end to which the program attaches the next track. You choose it by clicking an orange end.
Radius	How tight a curve is, measured from the centre of the circle to the middle of the track. Small radii save space; large ones look better, and long coaches run more safely on them.
Baseboard	The surface the layout is built on, often a wooden board, for example 2 × 1 m.
Millimetres	All dimensions in the program are millimetres: 1 m = 1000 mm, 1 cm = 10 mm. A 2 × 1 m board is therefore 2000 × 1000 mm.

More words are explained in the Glossary at the end of the manual.

Mouse and keyboard: what you need

Action	How to do it
Click	Press the **left** mouse button once briefly. This selects tracks and presses buttons.
Double-click	Click the left button twice quickly. On a track it opens its properties; on an empty area it shows the whole layout.
Right-click	Press the **right** mouse button once. A small menu opens with commands for whatever is under the mouse.
Drag	Hold down the left mouse button, move the mouse, then release. This moves tracks.
Move the view	Hold down the **right** mouse button and move the mouse. The plan moves with it.
Zoom in and out	Turn the **mouse wheel**. The `0` (zero) key shows the whole layout again.
Undo	`Ctrl` + `Z` takes back the last step, as often as you like. `Ctrl` + `Y` brings it back.
Cancel	`Esc` (top left of the keyboard) ends a tool or closes a window without changes.
Delete	Click the track and press `Del`.

> **Don't be afraid of mistakes**
> You cannot break anything. Every step can be taken back with `Ctrl` + `Z`. The **history** even jumps back up to 20 steps at once (on Windows **Home › History…**, in the web version the clock icon at the top right). Feel free to try everything.

Where is what?

The window always has the same layout:

- At the **top** is the **ribbon** with all commands. It is divided into **tabs**: **Home**, **Tracks**, **Scenery**, **Scenery items**, **Room**, **Wiring**, **View** and **3D**. Clicking a tab shows its buttons. At the far left is **File** with New, Open, Save and Print.
- On the **left** is the **track library** with all pieces of the chosen track system.
- In the **middle** is the **drawing area** with your plan.
- On the **right** are the **properties**. They show everything about what you have just clicked, and that is where you change values.
- At the **bottom** are the **level bar** and the **status bar** with the number of tracks, track length and open ends.

Move the mouse over a button and wait a moment. A small hint (tooltip) explains what it does. More about the window is in chapter 5.

Your first layout in 10 steps

We build a simple **oval** with a **siding** on a 2 × 1 m board in H0. It takes about a quarter of an hour. An oval consists of two semicircles of curves, joined by straight pieces. The train runs round and round on it.

1. **Create a new plan.** Click **File** and then **New** (or press Ctrl + N). The **New plan** window opens.
2. **Choose gauge and baseboard.** Under **Gauge** choose **H0** and as the shape **Rectangle**. Enter **2000** as the width and **1000** as the depth (millimetres!). Leave **Construction** at **Plan directly on level 1**. Click **Create plan** (web version: **Create**). An empty baseboard now lies on the drawing area.
3. **Choose a track system.** At the top of the track library choose your track system under **Manufacturer / track system**. The list below shows all pieces with article number and size. The buttons **Straight**, **Curves**, **Turnouts** etc. show only that kind.
4. **Place the first curve.** Click **Curves** and then once on a curve with a small radius. It appears on the baseboard. Its free end is **orange**: that is where the next track goes.
5. **Complete the semicircle.** Keep clicking **the same curve**. Each click adds a piece to the orange end. A semicircle is 180°. With 30° curves you therefore need six pieces; you can see the angle next to the track in the library. It is quicker with the Enter key: it adds the last chosen track again each time.
6. **Add straight pieces.** Click **Straight** and three or four times on the same straight. The line continues straight ahead. Remember how many straights you used.
7. **Build the second half with a turnout.** Add six curves again. Then click **Turnouts** and once on a turnout. It is added like a straight piece; its branch stays open as an orange end. Now, if necessary, click the orange end that continues **straight on**, and add straights until you are back at the start. If the ends meet exactly, the program joins them by itself. If a small gap remains, click one of the two ends and then **Tracks › Close gap**: the program inserts a matching piece of flexible track.

8. **Lay the siding.** Click the orange end at the **branch** of the turnout. Add two or three straights and finally, under **Other**, a **buffer stop**. If the branch points the wrong way, for example off the baseboard, select the turnout and press F (mirror). The status bar at the bottom then shows **open ends 0**.
9. **Check.** Press F7 (on Windows **Home › Collision check**, in the web version **Home › Plan check**). The list reports whether tracks touch, stick out over the edge or have open ends. Clicking an entry shows the spot. If nothing is listed, everything is fine.
10. **Look and save.** Click **3D › 3D view** and turn the layout with the left mouse button held down. Close the window and save with Ctrl+S. Enter a file name, for example “My first layout”. Done!

> **Tip**
> If a track faces the wrong way, press A (switch connection) or F (mirror) straight afterwards. This turns the last placed track around. If a track lies off to the side, drag it with the mouse to an orange end: it snaps into place by itself.

> **Example: Does the oval fit on the board?**
> A curve with a 360 mm radius needs 2 × 360 = 720 mm of depth for a semicircle, plus about the width of the track. On a board 1000 mm deep that leaves roughly 120 mm of margin at the top and bottom. With a 437 mm radius curve it would be 874 mm plus track, which is already tight. The plan check reports it if a track sticks out over the edge.

If something goes wrong

What happens	What helps
The track appears in the wrong place.	First click the orange end where you want to continue, then click the track in the library.
I can't see my plan any more.	Press 0. The whole layout is shown again.
A wrong track was placed.	Press Ctrl+Z, or click the track and press Del.
The oval does not quite close.	Click one end and choose **Tracks › Close gap** (chapter 8).
Every click places something instead of only selecting.	A tool is still switched on. Press Esc. After that a click only selects again.
A track cannot be moved.	It is locked. Untick **locked** in the properties.
Tracks have suddenly disappeared.	Their level is hidden. Tick it in the level bar at the bottom or click **Show all**.

What next?

Read the other chapters when you need a function:

- An L- or U-shaped baseboard, modules or the wooden benchwork: chapter 7.

- More about track, flexible track, parallel tracks and aligning: chapter 8.
- Gradients, a second level or a helix: chapter 9.
- Mountains, lakes, houses and trees: chapters 10 and 11. Fitting the layout into your room: chapter 12.
- Power, wires, signals and digital control: chapter 13.
- Shopping list with prices, full-size printout: chapter 16.

The **sample layout** “Newtown station” (on Windows **File › Load sample plan**, in the web version **File › Sample layout**) shows a finished layout with station, scenery, wiring and signals. Click individual parts: the properties on the right reveal how everything is done.

5 The workspace

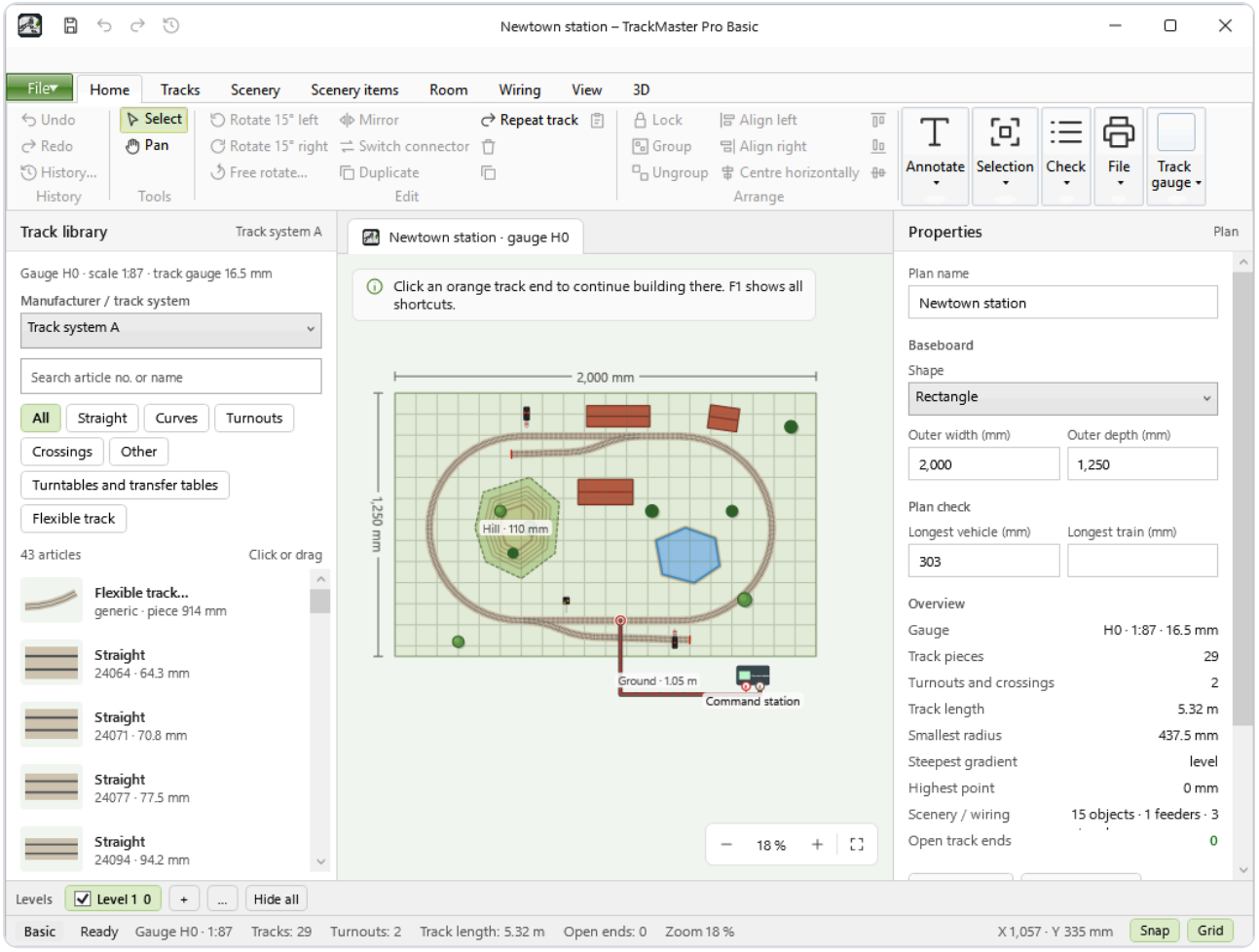


Figure 3 – The main window with the sample layout “Newtown station”.

The main window is divided into six areas:

Area	Purpose
Ribbon (top)	All commands, arranged in the tabs Home, Tracks, Scenery, Scenery items, Room, Wiring, View and 3D. Under **File** are New, Open, Save, parts list, exports, printing, sample, licence, appearance and language. The tabs are divided into **groups**; the group name is below its buttons.
Track library (left)	Choose the track system, search for track and filter by kind (All, Straight, Curves, Turnouts, Crossings, Other, Turntables and transfer tables, Flexible track). A click adds the track; dragging places it freely.
Drawing area (middle)	The track plan on the baseboard, with dimension lines at the edge, grid and the open track ends. At the bottom right are buttons for zooming in, zooming out and fitting.

Properties (right)	Show what is currently selected. With nothing selected you see the plan with its name, baseboard size and an overview. For a track you see position, height, usable length and digital address; for an object its dimensions. Values can be entered directly.
Level bar (bottom)	All levels of the plan. The tick shows or hides a level, a click on the name makes it the active level, **+** adds a new one above, **…** opens the Levels window. **Hide all** or **Show all** switches all of them at once (chapter 9).
Status bar (very bottom)	Gauge, number of tracks and turnouts, track length, open ends, mouse position in millimetres and zoom; plus the switches for **Snap** and **Grid**.

While a tool is switched on, a **tool bar** appears above the drawing area. It tells you what to do next and has buttons such as **Done**, **Undo point** or **Exit**. Short messages from the program appear at the bottom left and disappear by themselves after a few seconds.

Moving the view

- **Zoom:** mouse wheel, the `+` and `−` keys or the buttons at the bottom right.
- **Pan:** hold down the right or middle mouse button and drag. Or hold down the `Space` bar and drag with the left mouse button. Or switch on the **Home › Pan** tool.
- **Show the whole layout:** the `0` key, **View › Whole layout** or double-click an empty spot.

Rotating the view freely

You can turn the track plan on screen to any angle, for example the way you will later stand in front of the layout. Only the **view** is rotated: tracks, dimensions, parts list and printout stay unchanged.

- **With the mouse:** hold down `Ctrl` and turn the mouse wheel. This rotates in 5° steps, with `Shift` as well in 1° steps.
- **On the ribbon: View › 15° left**, **15° right** and **Rotate view…** for an exact angle.
- **Back:** while the view is rotated, a compass at the top right shows north and the angle. Clicking it or **View › Straighten** straightens the view again.

The dimension lines on the baseboard rotate too; their labels always stay readable. If you want to turn the **layout itself**, that is, really turn baseboard and track by 90°, use **Scenery › Rotate layout 90°** (chapter 7).

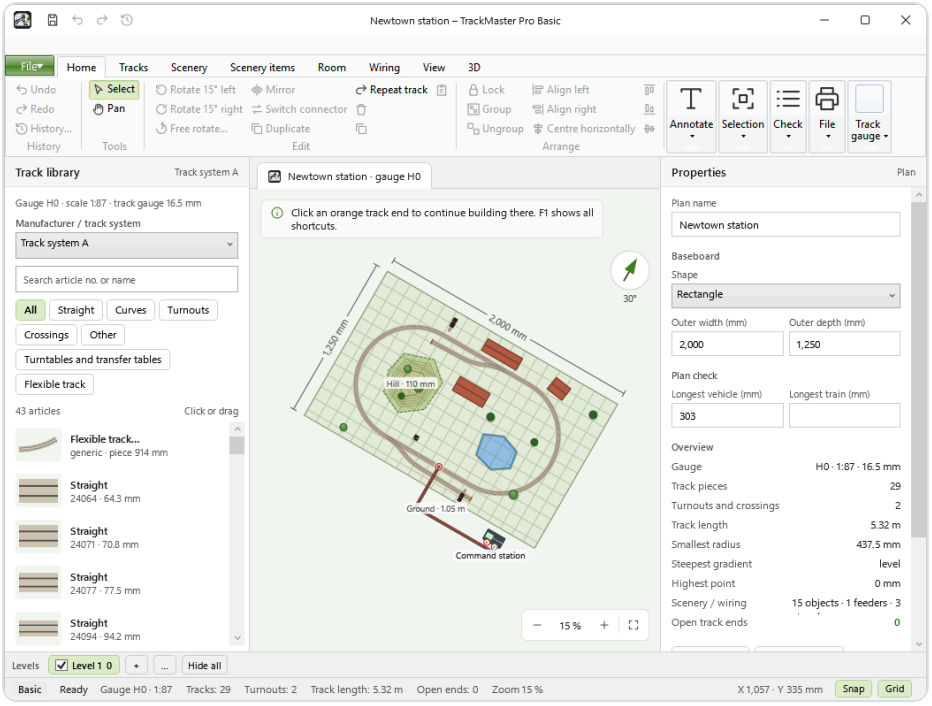


Figure 4 – View rotated by 30° with the compass at the top right.

Design and font

Under **View › Design and font** (also **File › Appearance…**) you choose how the program looks. The choice is saved and also applies at the next start. After **Apply** the window rebuilds briefly; your plan stays open.

Design	Appearance
Logo green	Default: green and anthracite, light working area.
Classic blue	Blue accents, light working area.
Blueprint	The plan as a technical drawing: white tracks on a blue background.
High contrast	Black tracks on a white baseboard, easy to read even in bright light or with weaker eyesight.
Workshop	Warm wood tones: baseboard like plywood, accents in copper.

Seven **fonts** are available, from the standard font through a narrow technical font to a serif font and a typewriter font. The **font size** has four levels: Small, Normal, Large and Very large. If you find screen text hard to read, choose **Large** or **Very large**. A preview in the window shows the effect immediately.

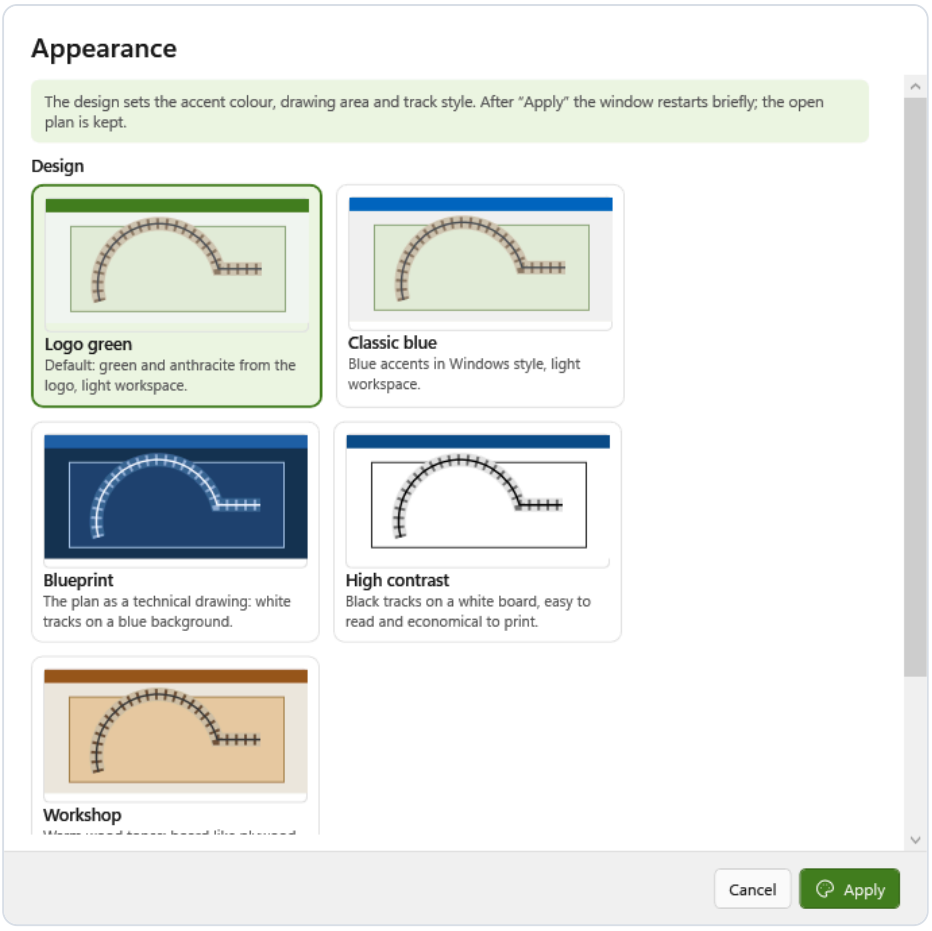


Figure 5 – Appearance window with designs, font and font size.

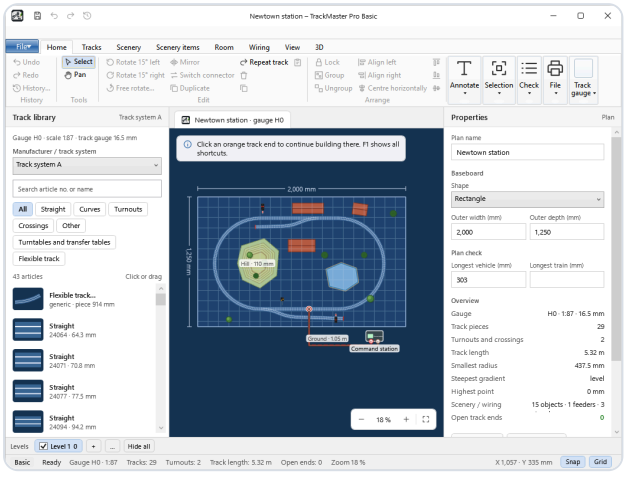


Figure 6 – Blueprint design.

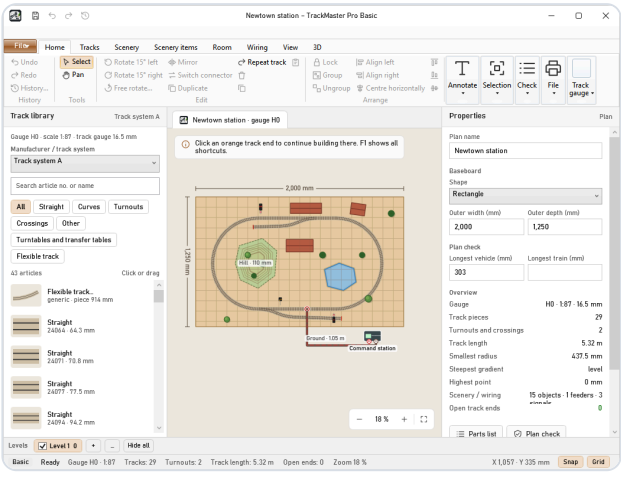


Figure 7 – Workshop design with a narrow font.

Language

The interface is available in German and English. At the first start it follows the system language. You can change it under **File › Sprache / Language...**. On Windows and in the browser the window then rebuilds and your plan is kept. On Android the new language applies from the next start of the app. Plans are independent of the language.

The View tab

In the **View** tab you switch on what the drawing area shows:

Switch	Effect
Grid	squares in the background for estimating distances
Track ends	marks the ends of all tracks, open ones in orange
Article numbers	writes each track's article number next to it, handy when building
Heights	writes the rail-top height at the track ends
Scenery	shows or hides hills, water, buildings and trees
Snap (web version)	switches track snapping on or off (on Windows: **Snap** in the status bar)
Levels...	opens the Levels window

In addition there are the groups **Rotate** and **Appearance** (see above) and **Help**: **Shortcuts** or **How to use** (F1) shows all mouse and keyboard commands at a glance, **About** shows the version and edition.

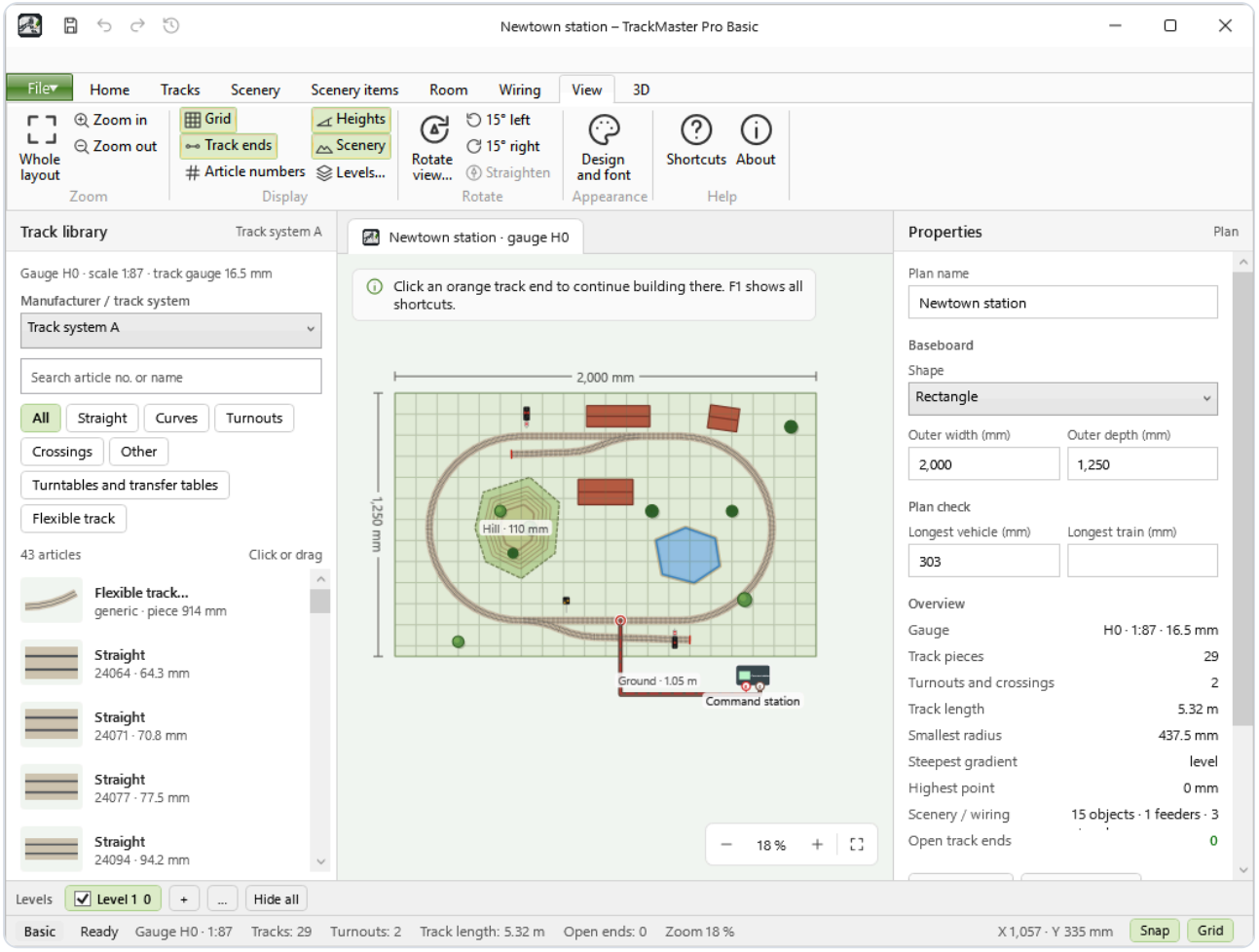


Figure 8 – The View tab.

6 Creating a plan and adding track

This chapter sums up the steps every plan starts with: creating a new plan, choosing the baseboard and track system, adding track. A detailed guide for your very first plan is in chapter 4.

Creating a new plan

1. Choose **File › New** (Ctrl+N). A plan that has not been saved is discarded; Ctrl+Z brings it back.
2. Choose the **gauge**, for example H0.
3. Choose the **baseboard shape** and enter the dimensions. The preview shows the shape immediately.
4. Choose the **construction** (see below). For a start, **Plan directly on level 1** is right.
5. Click **Create plan** (web version: **Create**).
6. Click an empty spot next to the baseboard and enter a name under **Plan name** in the properties on the right. It appears on the printout and is suggested as the file name when saving.

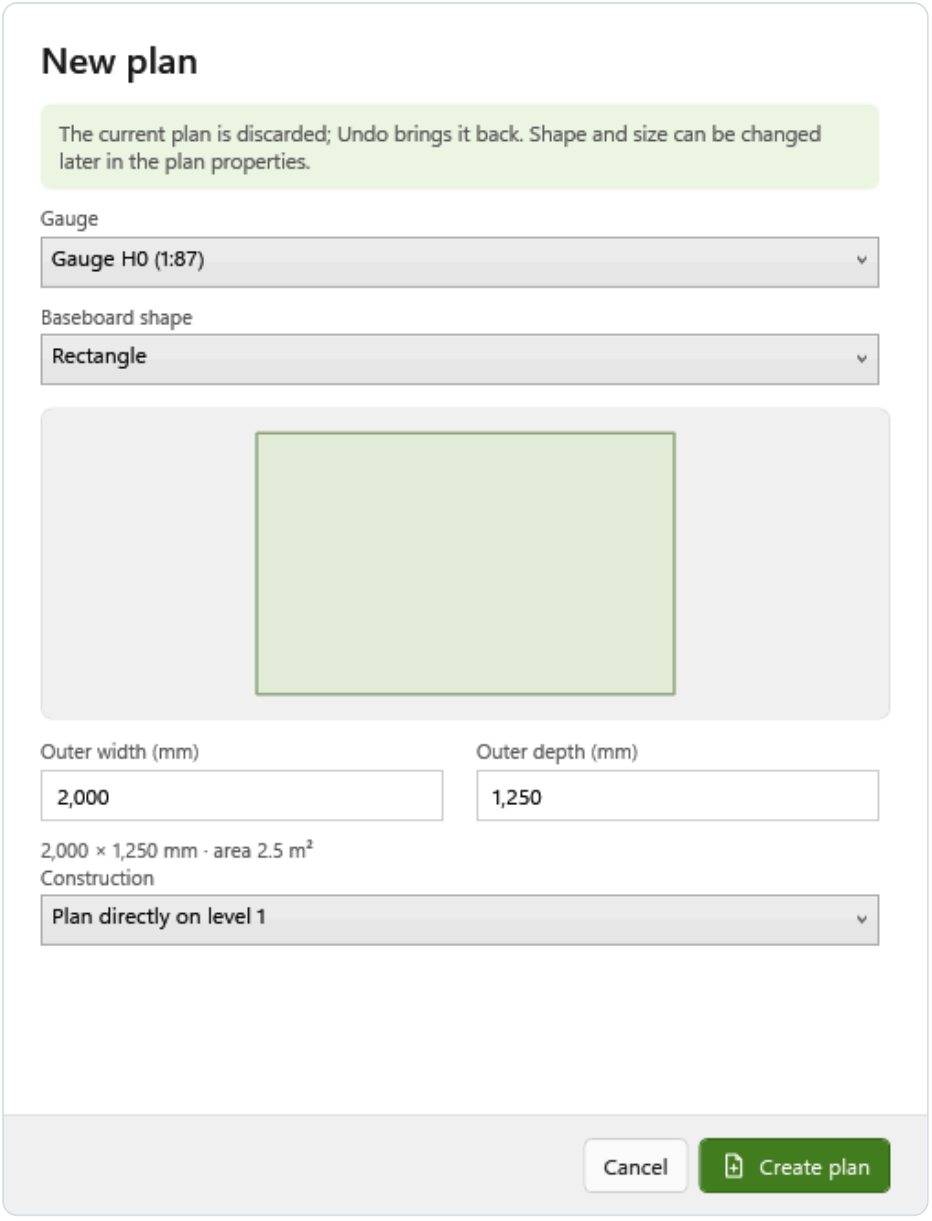


Figure 9 – New plan: gauge, baseboard shape, dimensions and construction.

Baseboard shape	Dimensions and special features
Rectangle	Width and depth. The simplest and most common shape.
L shape	Outer dimensions and leg depth, orientation in 90° steps. Good for a room corner.
U shape	Outer dimensions and leg depth; the two legs may be of different length. Good for three walls, with room to stand in the middle.
T and Z shape	Outer dimensions, leg depth and orientation.
E and G shape	Outer dimensions, leg depth and orientation. The E shape has three legs, the G shape an inward hook.
Ring	Layout all round with an open centre, for example with an access point in the middle.

Modules	Layout made of individual rectangular parts. Width and depth apply to the first module at first (chapter 7).

Construction. This decides whether a wooden frame goes under the baseboard right away:

- **Plan directly on level 1**: only the baseboard. You can add benchwork later at any time.
- **Benchwork from a template**: the plan gets a wooden frame immediately. You can choose **Frame with top board** (the usual layout table), **Open frame (rib construction)** without a top board, **L-girder frame** for large free-standing layouts and **Exhibition module (1,300 mm high)**.
- **Draw benchwork yourself**: only the baseboard is created and the **Scenery › Draw batten** tool is switched on straight away. You then draw the side rails, cross members, diagonals and braces yourself.

You can change the shape and dimensions later at any time: click an empty spot or **Scenery › Baseboard**; the properties then show the baseboard dimensions. More about baseboard, modules and benchwork is in chapter 7.

Choosing a track system

You choose the **gauge** in the new plan or later on the **Home** tab at the far right in the **Track gauge** group. You choose the **track system** at the top of the track library under **Manufacturer / track system**. The library then shows all pieces of that system with article number and size. You may also mix several systems in one plan, for example for a siding made of leftover stock; you then have to check yourself whether the transitions fit in reality.

> **Note on dimensions**
> The track geometry follows the catalogue data. A few track systems are stored with provisional dimensions and are marked in the library. Check these against the real track before building.

Adding track

The basic principle is always the same: there is an **active track end** (orange), and every track you click in the library is added there.

1. Click a track in the library. If the plan is still empty, it appears in the middle of the baseboard. Its free end is marked orange.
2. Click more tracks. Each new track is added at the active end. Enter or the Space bar adds the last chosen track again.
3. To continue somewhere else, click an orange end there. The E key jumps to the next open end.
4. If an end lands exactly on another, TrackMaster Pro joins the two by itself. An oval closes this way without any action from you.

Instead of clicking, you can also **drag** a track from the library onto the baseboard. Release it near an open end and it snaps into place there.

> **Tip: turnout the wrong way round?**
> A (switch connector) places the last track with its next connector at the same spot. This turns a turnout that splits the line into one that joins two tracks again. F (mirror) turns a left turnout into a right turnout, or makes a curve bend the other way.

Checking the plan

Press F7 now and then (Windows: **Home › Collision check**, web version: **Home › Plan check**). The list shows overlaps, tracks too close together, open ends and more. Clicking an entry shows the spot in the plan (chapter 14).

Sample layout

On Windows **File › Load sample plan**, in the web version **File › Sample layout** opens the layout “Newtown station” with scenery, wiring and signals. It is ideal for trying out all functions safely. Save your own plan first.

7 Baseboard, modules and benchwork

The baseboard can be dragged with the mouse or typed in to the millimetre, divided into modules, rotated and given a wooden frame.

Dragging the baseboard and typing dimensions

Switch on **Scenery › Drag baseboard**. The baseboard then shows handles:

- **Dragging corners** changes width and depth. Baseboards and modules are at least 50 × 50 mm.
- **Round handles** exist for L, U, T, Z, E, G shapes and the ring. On the inner edge, dragging changes the depth of the leg; at the free end, its length.
- **Dragging a module** moves it together with its track and scenery. Its edges snap to the neighbouring modules.
- **Clicking without dragging** opens the dimensions for typing: for a module its name, position (X, Y of the top left corner), width and depth; otherwise width, depth and leg depth of the baseboard. Clicking a round handle opens **Individual legs…**.

Esc ends the tool. Every step can be taken back with Ctrl+Z. Without the tool you can also change the dimensions in the properties on the right when nothing is selected, or via **Scenery › Baseboard**.

Individual legs

For L, U, T, Z, E, G shapes and the ring, all legs have the same depth at first. Under **Individual legs…** (in the baseboard dimensions or in the baseboard properties) each leg gets its own **depth** and, where it makes sense, its own **length**. Empty fields use the common leg depth or the full length. **Reset all** restores the original state.

> **Example**
> An L-shaped layout along two walls: the long leg at the window wall should be 600 mm deep, the short one by the door only 400 mm, so that the door stays clear. Create the L shape with a leg depth of 600 and enter 400 for the short leg under **Individual legs…**.

Rotating the layout

Scenery › Rotate layout 90° turns the baseboard with track, scenery and objects by 90° clockwise. Turning four times gives the original state again. In the dimensions window of a module, **Rotate 90°** turns only that module and its contents. If you only want to rotate the **view**, see chapter 5; the 3D view rotates freely in all directions (chapter 15).

Layout made of modules

Modules are individual rectangular layout parts that can be built and transported separately. Under **Scenery › Modules…** you put the layout together from such parts. An existing baseboard is split into

modules, one per rectangle of the shape. The table shows each module's name, position (X, Y) and size in millimetres.

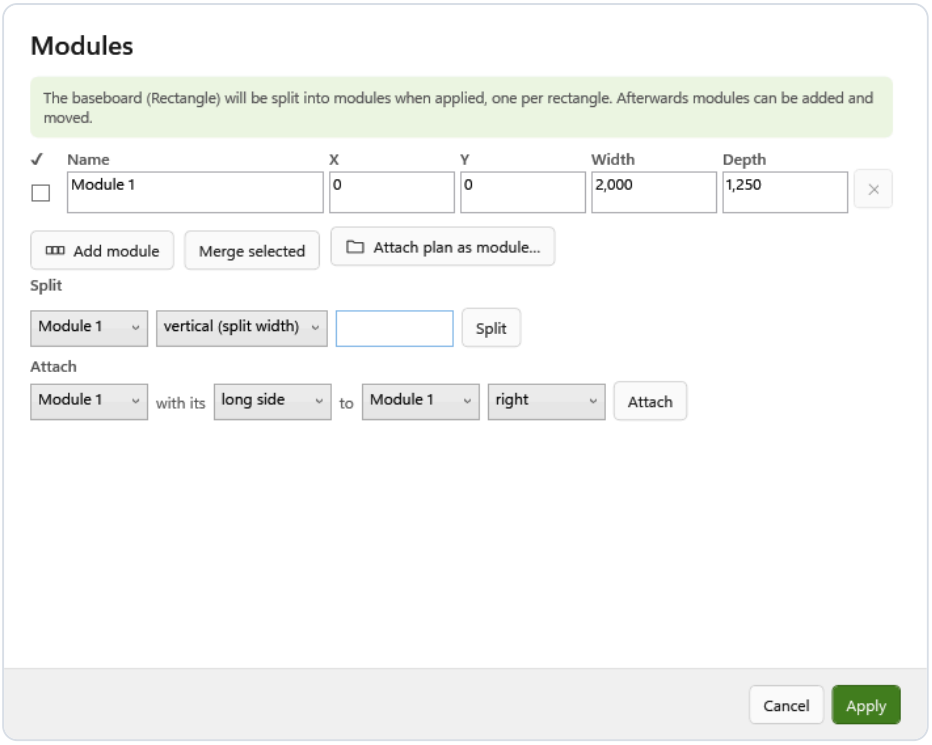


Figure 10 – Modules window with split and attach.

Command	Effect
Add module	new module to the right of the layout
Merge selected	joins the ticked modules into one, if together they form a rectangle
Split	divides a module **vertically** (splitting the width) or **horizontally** (splitting the depth) into two modules, at a position in mm or, with an empty field, in the middle. The parts are called "... a" and "... b".
Attach	places a module **with its long or short side** flush to the right, below, left or above another; it is turned by 90° if needed
Attach plan as module...	places a saved plan as a new module next to the layout, with all its track and scenery
×	removes the module

Only **Apply** changes the plan. Track and scenery move with their module. Track that runs across a module boundary is reported by the plan check, because that is where the layout will later be separated.

Web version only: module height and cutting at the edge

Height Z: in the **Height Z** column you set a module higher or lower than the others, for example a station module 80 mm above the line or a harbour basin 40 mm lower (negative value). The track on the module moves with it, new track lies on its surface, and the 3D view shows the module at its

height. Bridge the height difference to the neighbouring module with a gradient (**Tracks › Gradient…**); until then the plan check shows the transition as an open track end. In the **Thickness** column you set the thickness of the module board (empty = default).

Cutting track at the module edge: right-click a track that runs across a module edge and choose **Cut at module edge**. The program divides the track exactly at the edge into two pieces of flexible track. Position, height and gradient stay the same; feeders and isolating joints move with them. **Scenery › Cut at module edge** cuts the selected tracks, or with no selection all tracks across module edges. Turnouts, crossings and turntables cannot be cut; they must lie entirely on one module.

Wooden benchwork

A baseboard usually rests on a wooden frame. **Scenery › Benchwork…** plans this frame: side rails all round (the outer frame), cross members under each baseboard part, legs in the corners and along the long sides, and if you wish foot rails, levelling feet and a top board.

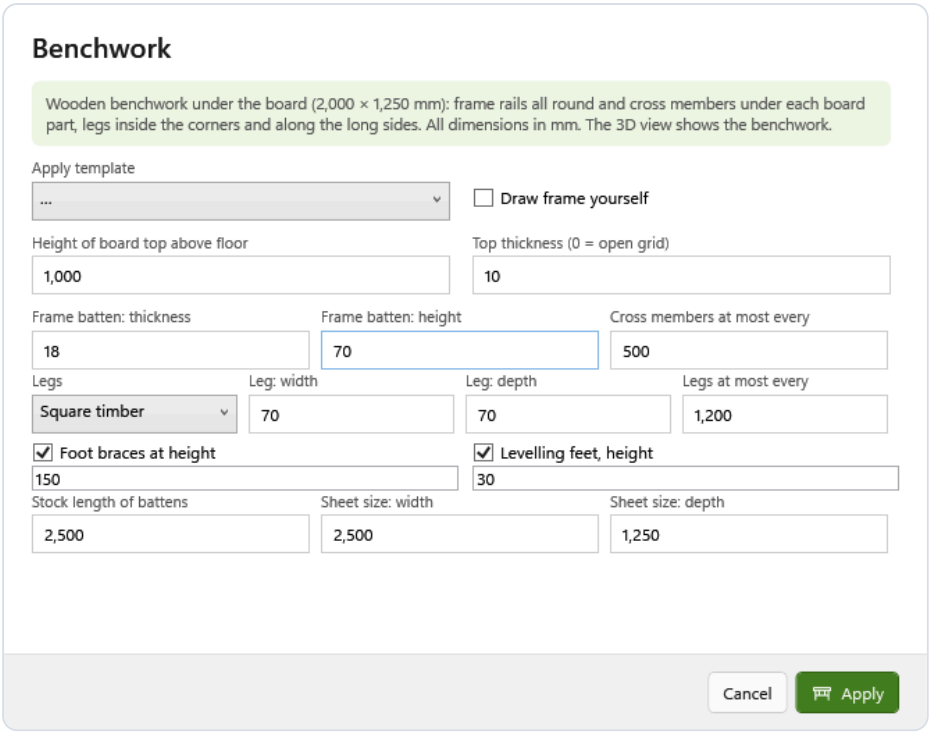


Figure 11 – Benchwork dimensions.

Setting	Meaning
Height of baseboard top above floor	Working height of the layout. 900 to 1100 mm is usual; less if you sit, more for exhibitions.
Board thickness	Thickness of the top board; 0 gives an open frame for rib construction
Batten for the frame	Thickness and height of the side rails and cross members; cross members at most every … mm

Legs	Square timber or L-leg made of two battens, width and depth, spacing at most ... mm
Foot rails, levelling feet	Height of the foot rails and the levelling feet (M8 with T-nut), so the layout stands level on an uneven floor
Batten length, board size	the lengths and board sizes available at the DIY store; the material list is based on them

The **material list** lists battens and square timber by stock length, boards, screws and glue: your shopping list for the DIY store. The **cutting list** shows every part with cross-section, length and quantity. Both can be copied or saved as a CSV file. Parts that are too long and top boards that are too big are reported with a note on joining them. The 3D view shows the benchwork under the baseboard; with an open frame (rib construction) it shows only the wooden framework with battens and track. **Remove** in the window takes the benchwork out of the plan again.

Diagonals and braces

If modules are staggered, for example in steps along a slanted wall, a continuous diagonal with individual braces often carries them. **Scenery › Draw batten** draws such battens freely: click the start and end. Baseboard corners, baseboard edges and other battens snap. If a brace starts at a module edge, its end snaps exactly onto the diagonal, vertically or horizontally below. The properties show the batten's width, height, length and top edge (negative = below the baseboard). The **arrow keys** move a selected batten by 1 mm, with Shift by 10 mm.

Laying a top board

Scenery › Lay top sheet lays a top board on an open or self-drawn frame. It gets the outer dimensions of the drawn battens (without battens, the baseboard dimensions) and rests on the highest batten. In the properties you can type the top left corner, width, depth, thickness and top edge to the millimetre. **Fit to frame** places the board exactly over the frame again.

8 Laying and editing track

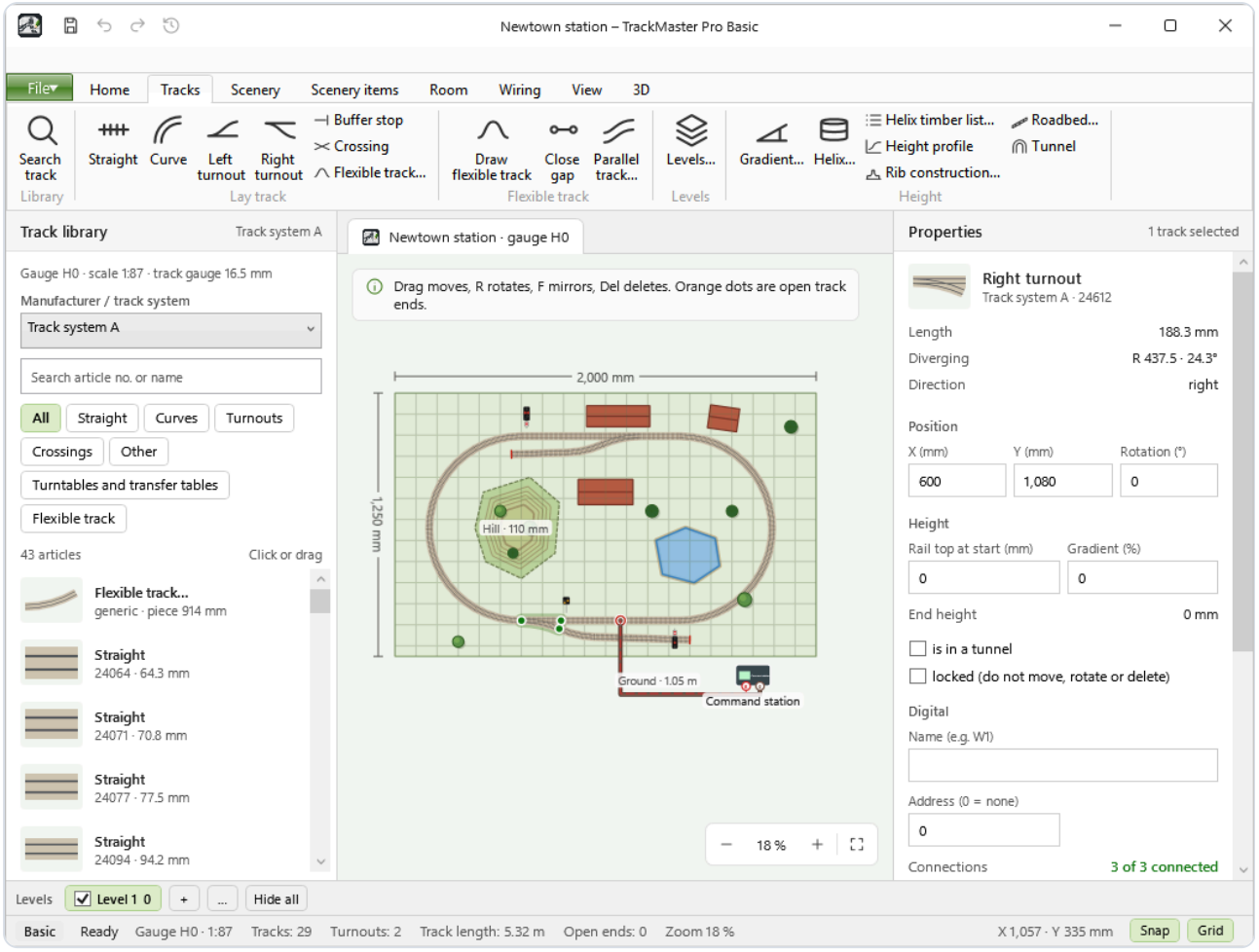


Figure 12 – The Tracks tab with a selected turnout; the properties show its dimensions, height and digital address.

Track library

- Choose the **track system** in the list at the top.
- **Search** by article number or name. **Tracks › Search track** (Ctrl + F) jumps straight to the search box.
- **Filter** by All, Straight, Curves, Turnouts, Crossings, Other, Turntables and transfer tables, and Flexible track.
- **Clicking** adds the track at the active end; **dragging** places it freely on the baseboard; near a free end it snaps into place.

The **Tracks › Lay track** group has quick commands for **straight**, **curve**, **left turnout**, **right turnout**, **buffer stop** and **crossing** of the chosen system, so you do not have to search the list first. On Windows it also has **Flexible track...**, in the web version **Turntable** and **Parallel track...**.

Turntables are available for every gauge and track system. They are in the library under the **Turntables and transfer tables** filter and are added like track; each track exit is a separate track end. Where the manufacturer offers no turntable of its own, a generic one is provided (“generic”). Its dimensions are provisional; check them before building.

Selecting and moving

- **Clicking** a track selects it. Shift +click adds more.
- **Dragging on an empty area** draws a selection frame: everything inside is selected. Ctrl + A selects everything.
- **Dragging** moves the selection. If a free end comes near another, the whole group snaps there. A single track snaps to any free end, even if it is the wrong way round; it is turned to fit.
- **Arrow keys** move by 1 mm, with Shift by 10 mm.
- Switch snapping off with **Snap** in the status bar if you want to place a track freely.

When a track is selected, the properties show, besides position, rotation and height, its **usable length**: the length of the track section up to the next turnout, crossing or track end. That is the maximum length of a train standing there. If a **longest train** is entered in the plan properties, the usable length turns red if it does not fit.

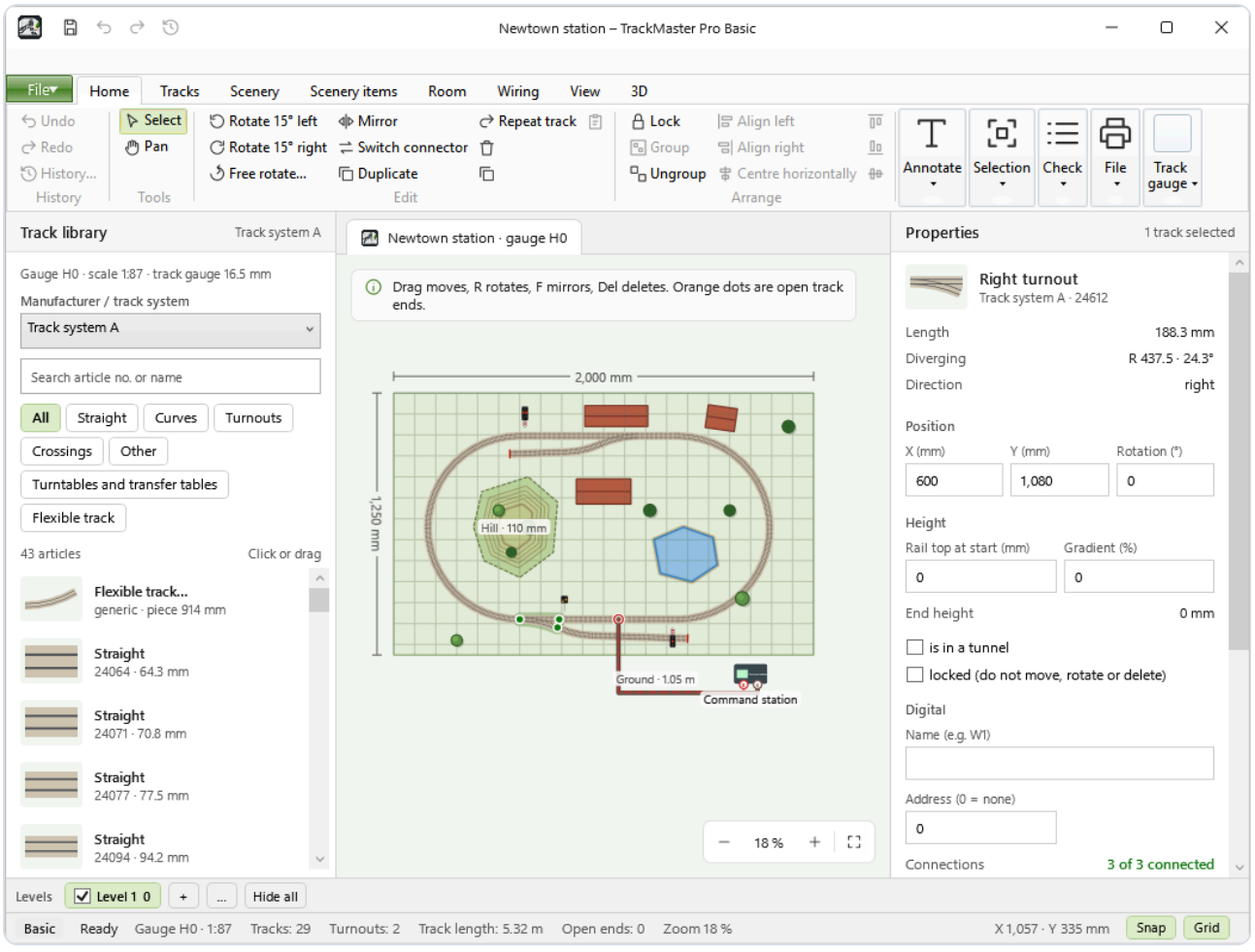


Figure 13 – The Home tab with the Arrange group; in the properties the “locked” tick box.

Editing

The commands are on the **Home** tab in the **Edit** group. Right-click a track to find the most important ones in the context menu as well.

Command	Key	Effect

Rotate 15° left / right	Shift + R / R	rotates the selection about its centre
Free rotate…	–	rotates the selection by any angle (quick choices ±15°, ±45°, ±90°)
Mirror	F	curves, turnouts and crossings reversed left to right
Switch connector	A	places the track with its next connector at the same spot
Repeat track	Enter or Space	adds the last track again at the active track end; a mirrored curve stays mirrored
Duplicate	Ctrl + D	copy of the selection next to it
Copy / Paste	Ctrl + C / Ctrl + V	track with feeders, isolating joints and signals via the clipboard, also into another plan of the same gauge; pasted at the mouse position
Delete	Del	removes tracks or the selected object
Undo / Redo	Ctrl + Z / Ctrl + Y	step by step back or forward
History	–	list of the last 20 steps; a click jumps straight to that state. Windows: **Home › History…**, web version: clock icon at the top right

The position (X, Y in millimetres) and rotation of a track can also be entered exactly in the properties.

Labels and dimension lines

In the **Home › Annotate** group you place **text** (click the spot and type the text) and **dimension lines** (click two points; track ends and baseboard corners snap). The program works out the length itself. Change text height, rotation and text in the properties. Labels and dimensions also appear on the printout.

Arrange: lock, group, align

The **Home › Arrange** group helps as the plan gets bigger:

Command	Effect
Lock	Selected tracks can no longer be moved, rotated or deleted. Ideal for finished areas that must not slip by accident. A second click unlocks them. You can also lock individual tracks with the **locked** tick box in the properties.
Group	Combines the selected tracks into a group. Clicking one then selects all, and they move together, for example a whole station. **Ungroup** separates them again.

Align	Aligns several separate track groups flush: left, right, top, bottom or centred horizontally or vertically. Select at least two groups that are not connected to each other.

When several tracks are selected, the same commands are also available as buttons in the properties. In the **Height of all selected tracks (mm)** field you put all selected tracks at a common height; any gradient is removed. For a ramp use **Gradient…** (chapter 9).

Parallel track

For a double-track line or a passing loop, TrackMaster Pro lays a second track next to existing track at a fixed distance:

1. Select the tracks of the line (draw a frame or Shift+click). They should be connected.
2. Click **Tracks › Parallel track…**.
3. Enter the **track spacing** from track centre to track centre. The default is the manufacturer's spacing for the chosen track system, or without such data the usual spacing for the gauge. The **Manufacturer track spacing** list inserts the value of another system.
4. Choose the **side**: left or right in the direction of travel, and confirm.

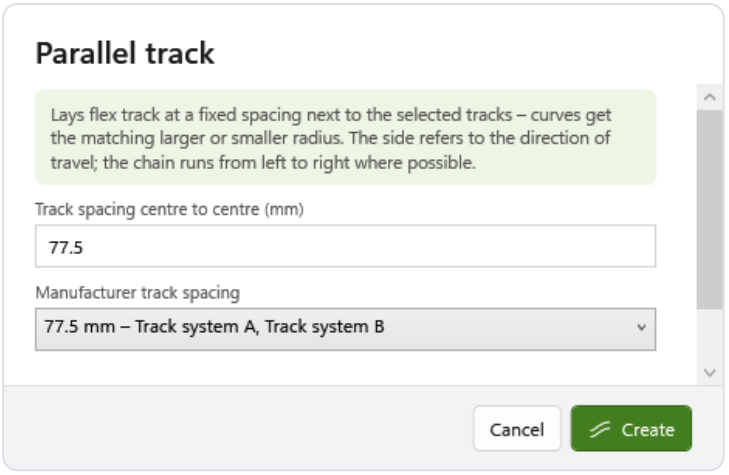


Figure 14 – Parallel track window.

The new track is laid as **flexible track** of the chosen system; curves automatically get the matching larger or smaller radius. If it is on the wrong side, undo it with Ctrl+Z and choose the other side.

Layout suggestion from your own track stock

You have a box full of track but no idea for a layout? The layout suggestion builds small layouts from your quantities, as ideas to get started. *This function is only available in the web version.*

1. Set the **baseboard** (size and shape) and choose your **track system** in the track library.
2. Click **Tracks › Layout suggestion…**.
3. For each track, enter **how many** you have. Empty or 0 means none. If you have already entered your stock in the parts list, it is already here.

4. Click **Calculate suggestions**. After a moment the suggestions appear with a small preview, number of tracks and size.
5. If you like one, click **Apply**. Ctrl + Z removes it again.

Depending on your stock, these shapes are possible: **oval**, **oval with siding** (or two), **oval with passing loop**, **double oval** (two loops inside each other) and the **figure of eight** with a crossing in the middle. Every suggestion is closed, fits on the baseboard and passes the plan check. It never uses more track than you entered.

> **No suggestion?**
> An oval needs curves for a full circle of one radius (for example 12 curves of 30°), and the baseboard must be at least twice as deep as the radius plus some margin. Sidings need a turnout, a matching opposite curve and a buffer stop. If “Replace existing tracks” is ticked, the suggestion replaces the track in the plan; otherwise it is added.

Flexible track

Flexible track comes in long lengths that you bend yourself and cut to length. With it you can lay lines that are impossible with fixed pieces. There are three ways:

Drawing flexible track

1. Switch on **Tracks › Draw flexible track**.
2. Put the mouse on an open track end and drag. The flexible track runs out of the track without a kink and bends as you move.
3. When you get close to another track end, it snaps. The flexible track is then bent so that it joins both ends without a kink.
4. Release the mouse button. Length and smallest radius are shown while you drag.

Reshaping: with the tool switched on, grab a laid flexible track at one end and drag; the other end stays in place.

Closing a gap

Click an open track end, if necessary select the opposite end, and click **Tracks › Close gap**. TrackMaster Pro joins both ends with a suitably bent piece of flexible track. Without an opposite end it takes the nearest open end.

Flexible track with dimensions

Via the “Flexible track…” entry in the library (on Windows also **Tracks › Flexible track…**) you create a straight piece of any length or a curve of any radius and angle.

If a track system has its own flexible track, it is listed separately in the library. The parts list counts flexible track in whole lengths, with 5 % waste.

Caution

When dragging, watch the smallest radius shown. It should not be smaller than the smallest curve radius your vehicles can manage. If your locomotive's instructions say nothing about it, the smallest curve of your track system is a good guide.

9 Heights, levels, helix and rib construction

Many layouts have more than one storey: a line crosses a bridge, a station is hidden under the scenery, or the railway climbs a mountain. This chapter explains how to raise track, build ramps, organise levels and have the timber parts for the benchwork calculated.

> **How is height measured?**
> All heights in the program are the height of the **rail top** above the baseboard, in millimetres. A track lying directly on the baseboard therefore has height 0. If it lies on a 10 mm cork strip, it is still 0 – you only enter something when it lies *higher than the other tracks*.

Height of a track

Every track has a **height at the start** and a **gradient** in percent. The program calculates the height at the other end itself. To change a track's height:

1. Click the track. Its **Properties** appear on the right.
2. Enter the height you want under **Rail top at start (mm)**, for example **80**.
3. If the track is to stay level, leave the gradient at 0 %.

If several tracks are selected, the height field puts **all** of them level at this height. With **View › Heights** the heights are written at the track ends in the plan. The program always draws higher lines over lower ones, so you can easily see bridges and underpasses.

Gradient over a run of track

A **ramp** is a stretch of line that rises evenly. Instead of calculating every track yourself, let the program spread the gradient.

1. Select all tracks of the ramp: draw a frame around them with the mouse, or hold down Shift and click the tracks one after another.
2. Click **Tracks › Gradient…**.
3. Enter the **start height**, that is, the height at the bottom of the ramp.
4. Under **Input**, choose what you know:
   - **End height** – you know how high the ramp must arrive at the top. The program calculates the gradient.
   - **Gradient in percent** – you want a particular gradient. The program calculates the end height.
5. If the ramp rises in the wrong direction, reverse the direction. Length, gradient and end height are shown below the fields. Confirm.

> **Example: How long must the ramp be?**
> One line is to cross over another. In H0 you need about 80 mm of height for this. At a 2.5 % gradient the track rises 25 mm per metre. For 80 mm you therefore need 80 ÷ 25 = 3.2 m of ramp. At 3 % it would be 2.7 m. Ramps are almost always longer than you think – plan them early.

If two parallel runs of track are selected, for example a double-track ramp, the inner, shorter track gets exactly the gradient entered. The outer one rises so that it arrives at the same height.

Gradient	Assessment
up to 2 %	very good, as on the real railway; every train gets up
2 to 3 %	usual on model railway layouts
over 3 %	the plan check warns; long or heavy trains need a second locomotive
over 5 %	the plan check reports an error; most locomotives slip here

Height at a joint *(web version only)*: double-click the point where two tracks meet – or right-click and **Height at joint…** – to change the height exactly there. Both tracks get the new height at this point, their other ends stay where they are, and the gradient adjusts. Turnouts and crossings are raised as a whole. With tracks selected, the heights at the joints are shown in the plan (↕ in mm).

Height profile

The height profile is a small chart that shows how the line goes up and down – like the profile of a mountain stage in a cycle race. Select connected tracks, for example a ramp or the whole line between two stations, and click **Tracks › Height profile**. The horizontal axis is the distance travelled, the vertical axis the height. **Red** parts are steeper than recommended; lengthen the ramp there.

Levels

Think of levels as the storeys of a house: at the bottom there may be a hidden **fiddle yard** (a storage yard under the layout where trains wait), above it the visible **main level** and at the top a **mountain line**. Every level has a name and a **base height**. Every track and every scenery item belongs to exactly one level. A layout can have up to **25 levels**.

The big advantage: you can **hide** a level. Then you see, for example, only the fiddle yard and can work there without the tracks above getting in the way.

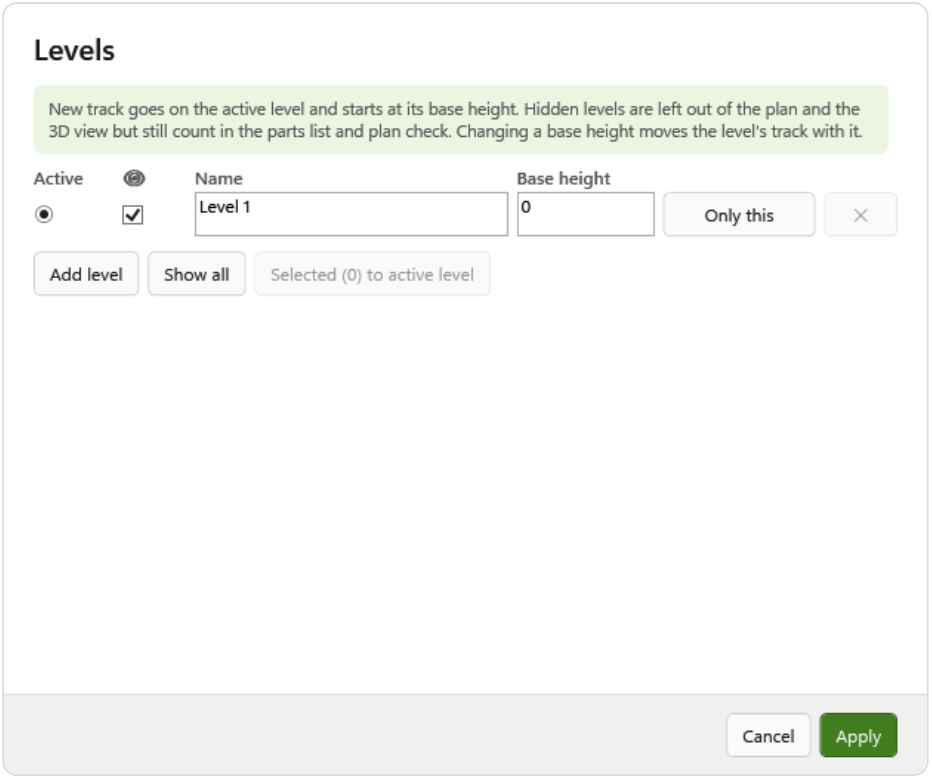


Figure 15 – Levels window.

The level bar

The level bar is at the bottom of the window. It has a button for each level.

- The **tick** next to the name shows or hides the level.
- **Clicking the name** makes the level *active*. New tracks always go onto the active level and start at its base height.
- **+** adds a new level above.
- **Hide all** hides all levels with one click; the same button is then called **Show all** and brings them back. Handy if you then want to switch just one level back on.
- **…** opens the Levels window.

The Levels window

Tracks › Levels… (or **View › Levels…**) opens a list of all levels. There you can

- **add** and **rename** levels and change their **base height**,
- **remove** a level – its tracks move to the level below, nothing is lost,
- show **Only this** level or bring all back with **Show all**,
- move selected tracks to another level with **Selected to active level**. Their height changes by the difference between the two levels.

If you change a level's base height, all its tracks move with it. Hills of a level stand on its base height, so the terrain adapts by itself.

Important
Hidden levels are only missing from the display (plan and 3D view). They still count in the parts list and the plan check. If tracks have “suddenly disappeared”, their level is almost always hidden.

Helix

A **helix** is a line that winds upwards in a circle like a spiral staircase. It overcomes large height differences in little space – typically to connect a fiddle yard under the layout with the main level. The program builds it from curved track of your track system: **round** or **oval**, **single-track** or **double-track**.

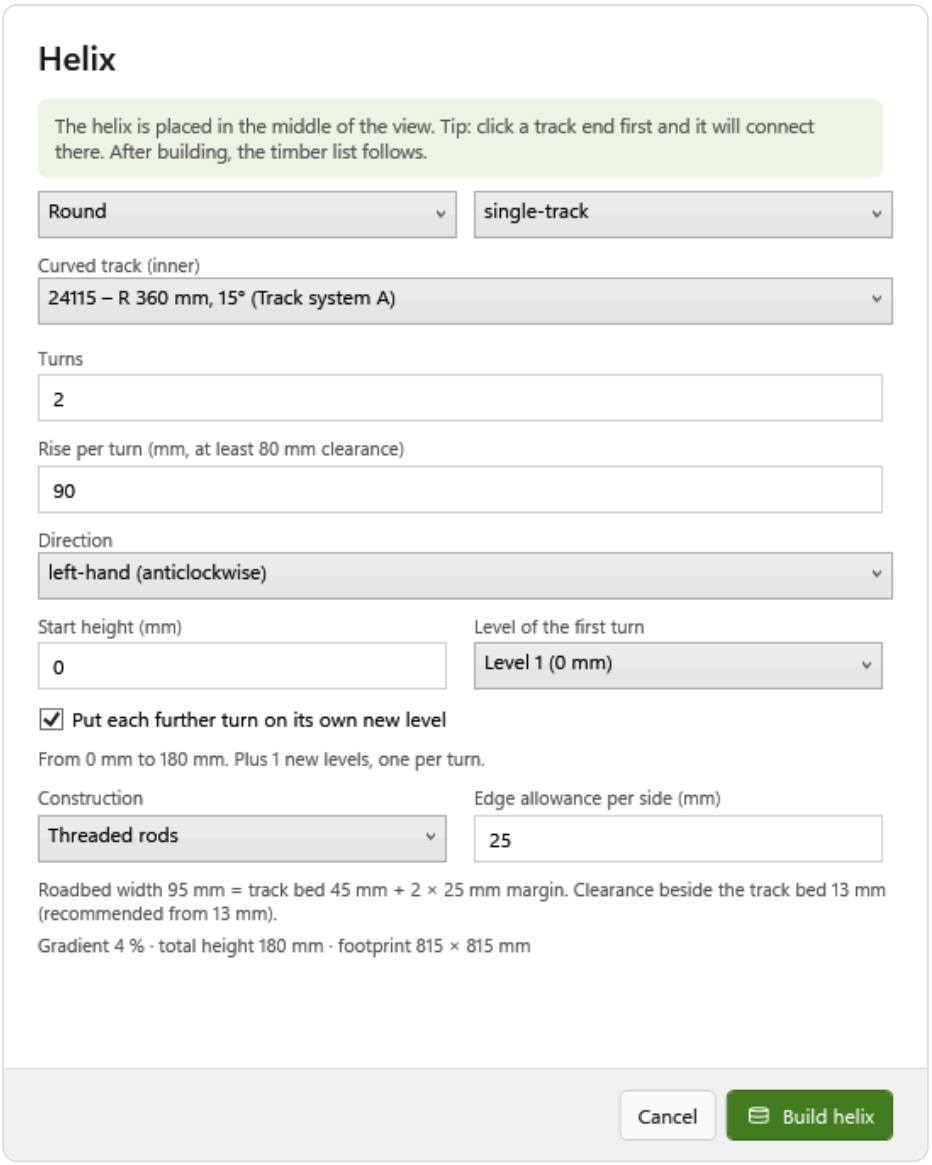


Figure 16 – Helix window with gradient, total height and footprint.

1. Click a free track end if the helix is to connect there. Without a selected end it is created in the middle of the view.
2. Click **Tracks › Helix…**.

3. Choose the **shape**: *Round* or *Oval*. In an oval there are straight tracks on both long sides; choose the straight and how many per side.
4. Choose *single-track* or *double-track*. With double track the second track is on the outside: as a curve with a matching radius or as flexible track with a freely chosen spacing.
5. Choose the **curve** for the inner track, the number of **turns**, the **height per turn** and the **direction** (left- or right-handed).
6. Under **Height and levels** are the **start height** and the **level of the first turn**. The defaults are the height of the selected track end or the base height of the active level.
7. With *Put each further turn on its own new level* (the default) every turn gets its own level: "Helix – turn 2", "… 3" and so on. This lets you show and hide the turns individually later.
8. Gradient, total height, footprint, start and end height are shown below the fields. Click **Build helix**.

> **Example**
> Curve with a 360 mm radius, height per turn 80 mm. One turn is 2 × 3.14 × 360 ≈ 2260 mm long. 80 mm over 2260 mm is a 3.5 % gradient – already steep. With the next larger radius (about 420 mm) it is only 3.0 %. Larger radii make the helix gentler but need more space.

The height per turn must be at least the **clearance height** of the gauge, otherwise no train fits under the next turn. The window points this out.

Under **Construction** you choose how the roadbed is supported: on **threaded rods** (long bolts on which the boards are held with nuts) or in **rib construction** (vertical boards with slots in which the roadbed boards lie).

Changing the helix later, inserting a turnout

The finished helix consists of individual tracks. Each can be changed like any other track. So that the other turns are not in the way, hide all levels in the level bar except the turn you want. To build a turnout into the helix:

1. Select the track that the turnout is to replace.
2. In the **Properties** under **Insert turnout**, choose the turnout and if needed **in the opposite direction** and **Branch on the other side**.
3. Click **Insert turnout**. The turnout takes over height, gradient and level. If it is longer than the track, the following tracks are replaced too; a gap is closed with flexible track. Ctrl + Z takes everything back.

> **Tip**
> In tight helix curves a **curved turnout** fits best because it is curved itself. A straight turnout often does not fit there – the program tells you.

Helix timber list PRO

After building, the program shows the **timber list**: everything you need from the DIY store for the helix benchwork, with finished cutting sizes. It can be copied or saved as a CSV file for a

spreadsheet.

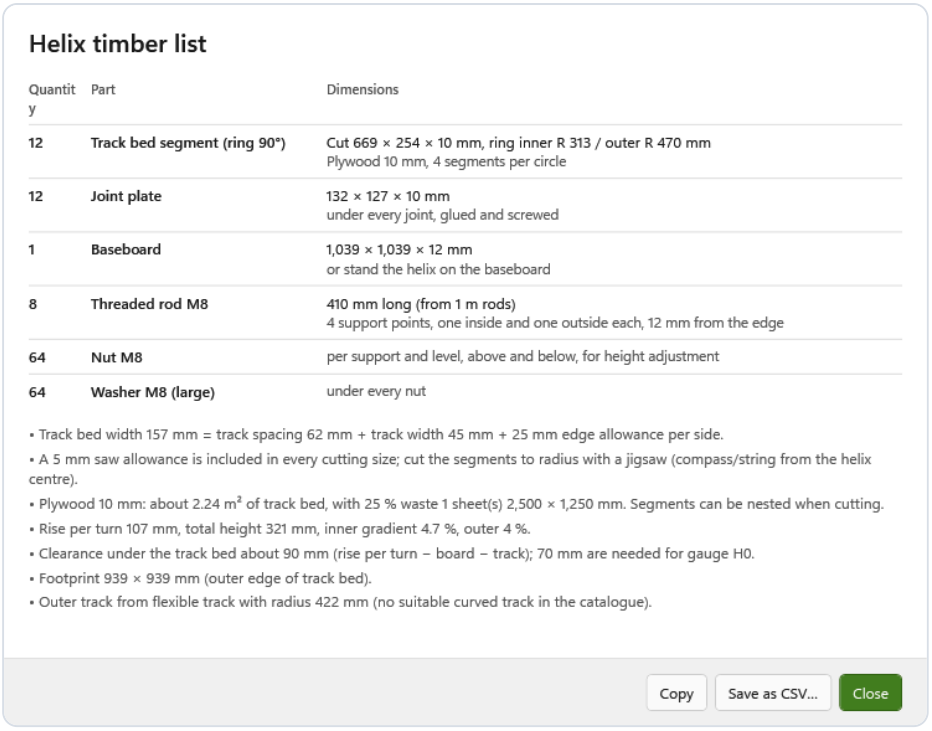


Figure 17 – Timber list of a double-track helix.

Part	What is it?
Roadbed segments	Curved plywood boards (pieces of a circular ring) on which the track lies. With cutting size and inner and outer radius. 4, 6, 8 or 12 per circle, so none is longer than 1200 mm.
Straight roadbed boards	in an oval, one board per long side and turn
Splice plates	small boards under each joint between two segments, for gluing and screwing
Base board	with a 50 mm margin all round – or stand the helix directly on the baseboard
Threaded rods, nuts, washers	one rod inside and outside per support point; nuts and washers above and below each turn so the height can be fine-tuned
Notes	roadbed width, board requirement, gradient inside and outside, free height under the roadbed and footprint

What the program calculates with

- **Track bed:** about 2.7 times the track gauge per track; with double track the track spacing is added.
- **Edge:** at least 25 mm per side – room for the threaded rod hole (12 mm from the edge) and a guard rail so no train falls off.
- **Saw allowance:** 5 mm, already included in every cutting size.

- **Timber thickness:** 8 mm for small gauges, 10 mm up to gauge 0, 15 mm for large gauges; base board at least 12 mm.
- **Threaded rods:** M8, M10 for large gauges.
- **Board requirement:** in boards of 2500 × 1250 mm with 25 % waste.

> **Tip**
> Mark out the segments from the centre with a string and a pencil (like a large compass) and cut them out with a jigsaw. Drill the holes for the threaded rods with a template so that all turns lie exactly on top of each other.

> **Caution**
> If the free height under the roadbed is tight, a warning appears in the list. Then increase the height per turn or use thinner timber.

Cutting template as SVG: this button in the timber list window saves every segment, board and splice plate as a separate drawing at full size (1:1), together in a ZIP file. Printed out, it serves as a template to stick on the wood; it is also suitable for plotters and CNC routers.

Rib construction for the layout PRO

In **open-frame construction** there is no continuous baseboard. Instead, vertical boards called **ribs** stand on the frame at regular intervals. Narrow roadbed boards lie on them only where tracks run. This makes it easy to form hills and valleys. **Tracks › Rib construction…** calculates the ribs from the track heights in your plan.

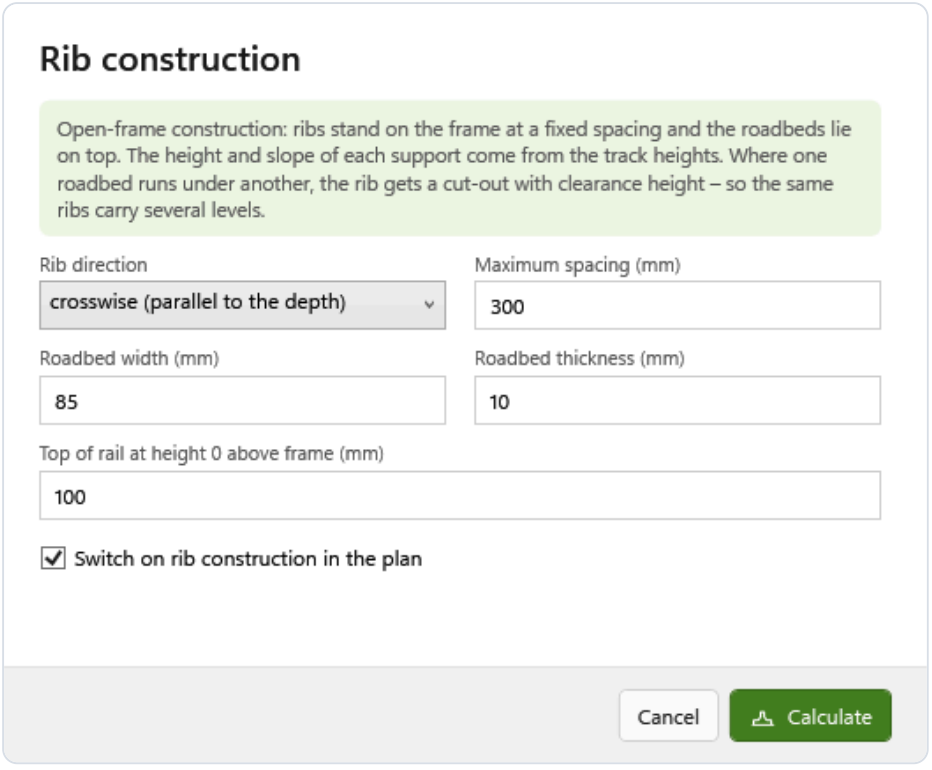


Figure 18 – Settings for rib construction.

- **Rib direction:** across or along the layout, with the maximum spacing in mm.
- **Roadbed width and thickness** and the **rail top at height 0** above the frame.
- Each support gets its height and slope from the track heights; beside the supports the top edge slopes away.
- If one roadbed lies below another, the rib gets a **cut-out** of clearance height – so the same ribs carry several storeys.

The result shows each rib with position, length, height, supports and cut-outs. It can be copied, saved as CSV and output as SVG: an overview and each rib as a 1:1 template, together in a ZIP file.

Roadbed from selection PRO

For a ramp or a freely routed line, **Tracks › Roadbed…** calculates the board under the selected tracks:

1. Select the tracks, for example a ramp.
2. Click **Tracks › Roadbed…** and enter roadbed width, thickness, maximum part length and splice plate length.
3. The roadbed follows the track centre and is divided into parts of equal length with splice plates. Save the parts as SVG: each part as a separate file at full size, with the track joints and their heights drawn in.

Tunnels

Select the tracks that are to be in the tunnel and click **Tracks › Tunnel** (also under **Scenery › Tunnel**). Tunnel tracks are drawn below the other tracks in the plan, get portals in the 3D view and may lie under a mountain in the plan check. Clicking **Tunnel** again with tunnel tracks selected turns them back into normal tracks.

10 Scenery

A model railway only comes to life with hills, water, houses and trees. On the **Scenery** tab you sketch the landscape roughly, so you can see how the layout will look – and so the plan check notices if a track runs straight through a mountain.

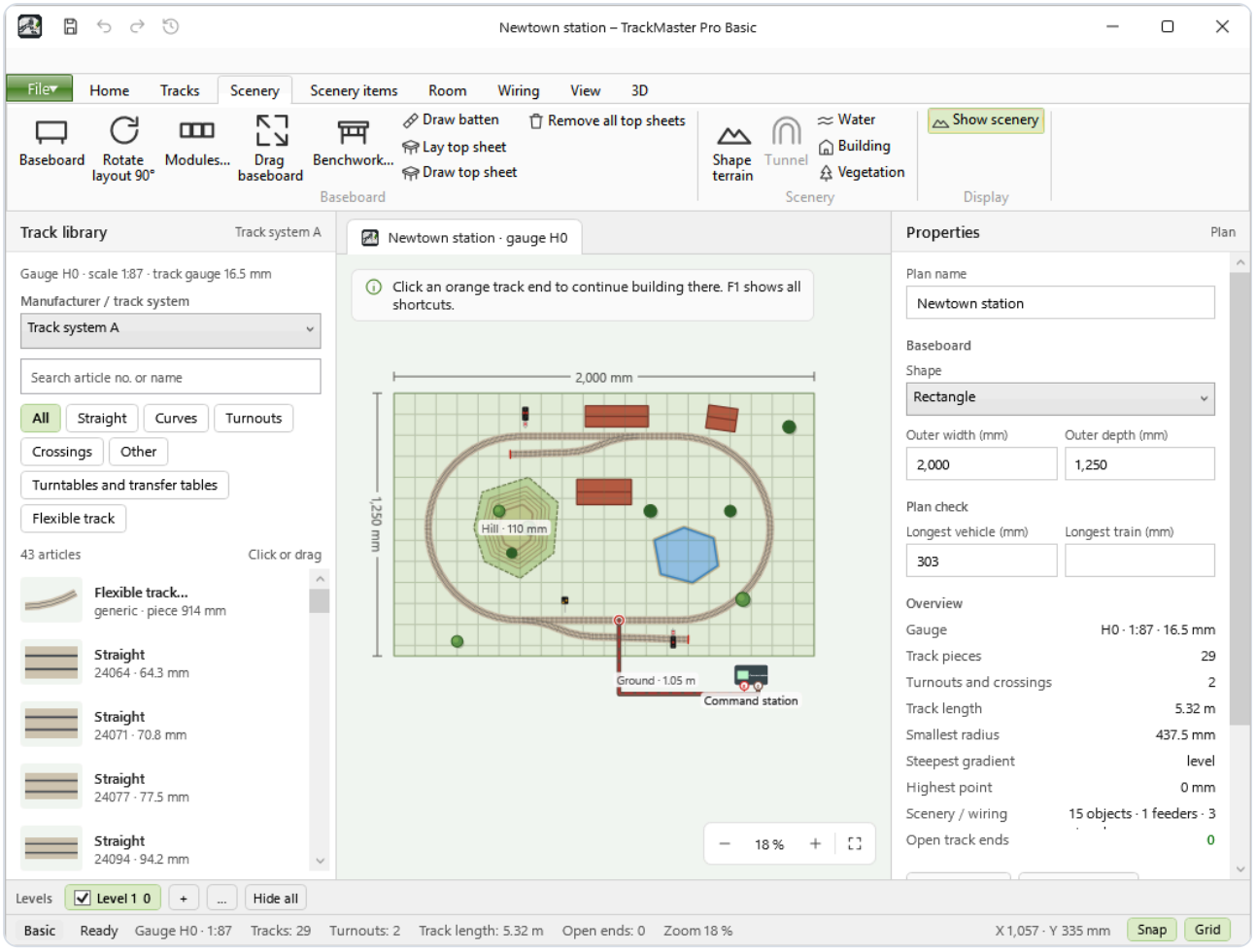


Figure 19 – The Scenery tab with hill, lake, buildings and trees.

Tool	How to use it
Shape terrain	Click the corner points of a hill, mountain or plateau one after another in the plan. Enter (or **Done**) closes the area, Esc cancels. Enter the height afterwards in the properties.
Tunnel	Turns selected tracks into tunnel tracks (see chapter 9).
Water	Draw a lake or river as an area, just like terrain. The depth is in the properties.
Building	Draw a rectangle with the mouse button held down. Width, depth, height and rotation can be entered exactly afterwards.
Vegetation	Every click places a tree. Diameter and height are adjustable.

Every object can be clicked, moved and deleted with Del. **Scenery › Show scenery** hides the whole landscape if you only want to see the track. The plan check reports track that lies in buildings, in water or under a mountain without a tunnel.

How the terrain is shown

- **Hills and mountains** rise from the edge of the area with a natural slope up to the height entered. In the plan, contour lines like on a hiking map show how steep the terrain is: the closer the lines, the steeper. Every fifth line is stronger.
- **In the 3D view** gentle slopes are meadow, steeper ones earth and very steep ones rock.
- **Water** has a bank strip. **Trees** are deciduous or conifers, with shadows in the plan.
- Buildings, trees and piers stand on the terrain – a tree on a hill stands on top of the slope.
- **Embankment:** if a track lies higher than the ground, it gets an embankment with a slope in the 3D view. Above other tracks no embankment is created, so a train fits underneath (there **Supports** in the 3D view shows piers).

11 Scenery items: basic models and your own models

You place tunnel portals, platforms, a station building, bridges and houses ready-made from the **Scenery items** tab. With Pro you also build your own buildings in the **model editor** – from simple shapes such as boxes and roofs.

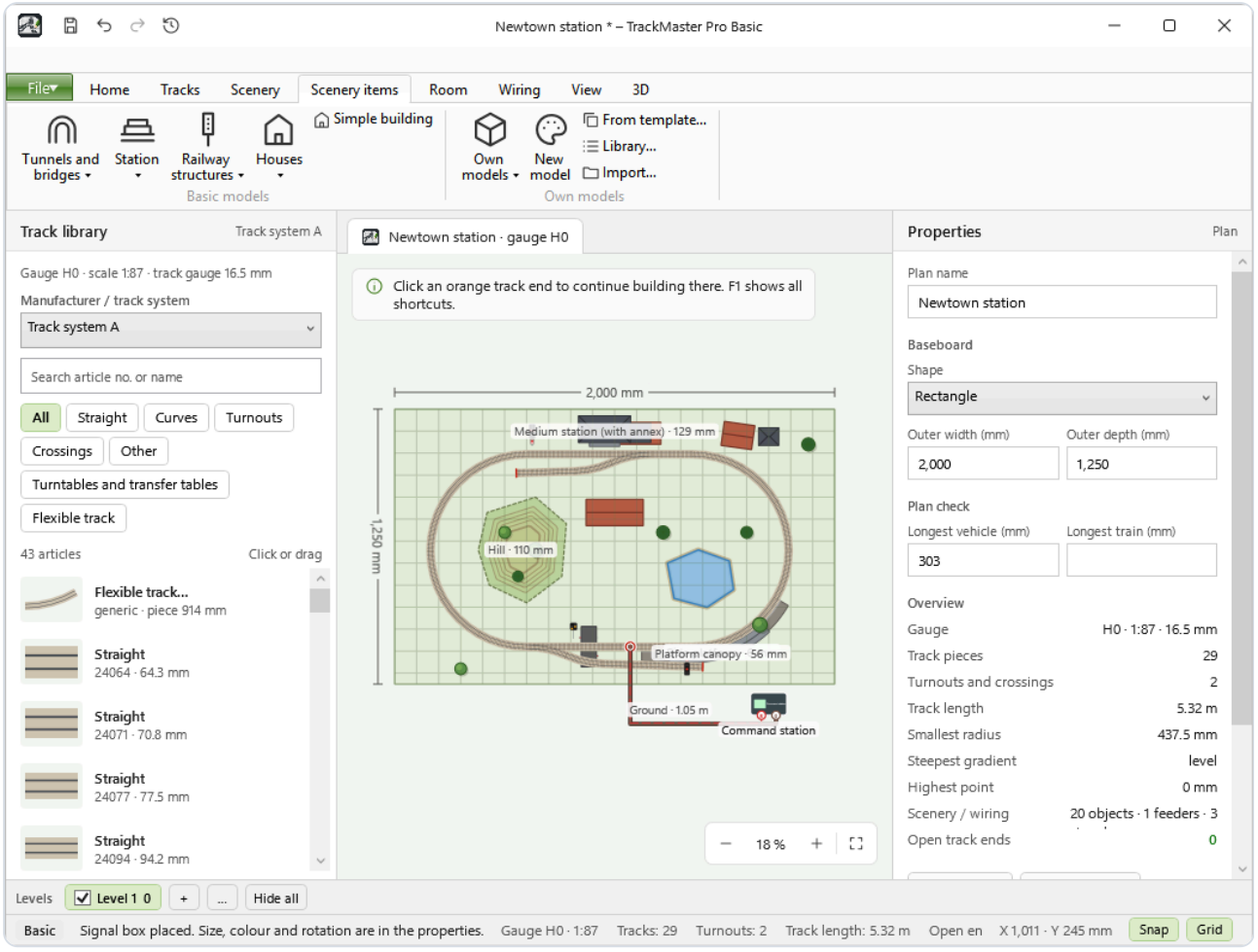


Figure 20 – The Scenery items tab with platform, platform canopy, tunnel portal, level crossing and station.

Inserting basic models

The basic models are part of the program and need no licence.

1. On the **Scenery items** tab, click a group, for example **Station**, and choose a model in it.
2. Click in the plan. The model appears at that spot.
3. You can place more copies. Esc or a second click on the same model ends the tool.

Group	Models
Tunnels and bridges	Tunnel portal single and double track, box girder bridge, bridge abutment, retaining wall

Station	Side platform, island platform, platform canopy, shelter, station building small and medium
Railway structures	Signal box, goods shed, engine shed single and double, level crossing, buffer stop with earth mound
Houses	Residential house, farmhouse, barn

The models have the size of the real prototype and are automatically scaled down to **your gauge's scale** (1:87 in H0). Many snap to the track by themselves:

- **Tunnel portals** sit across the track at the start of the tunnel.
- **Platforms and retaining walls** lie next to the track at the right distance and follow curves too. Your click decides which side they are on.
- **Bridges** go under the track, **engine sheds** over the track, the **level crossing** across it and the **buffer stop** at an open track end.

> **Placing along several tracks**
> First select the tracks along which a bridge, retaining wall or platform is to be built (draw a frame or Shift +click). Then choose the model: it is placed straight away along the whole selection, as long as the tracks and with all curves. **Switch side** in the properties puts a wall or platform on the other side, **Both sides** adds a second one opposite.

A model snapped to the track **remembers its tracks**. If you later raise the tracks or move them to another level, the model adapts by itself. The **Properties** show length, depth, height, rotation and a **main colour** (for example render in cream or brick red). **Original size** restores the size to scale. **Simple building** draws a plain house as a rectangle.

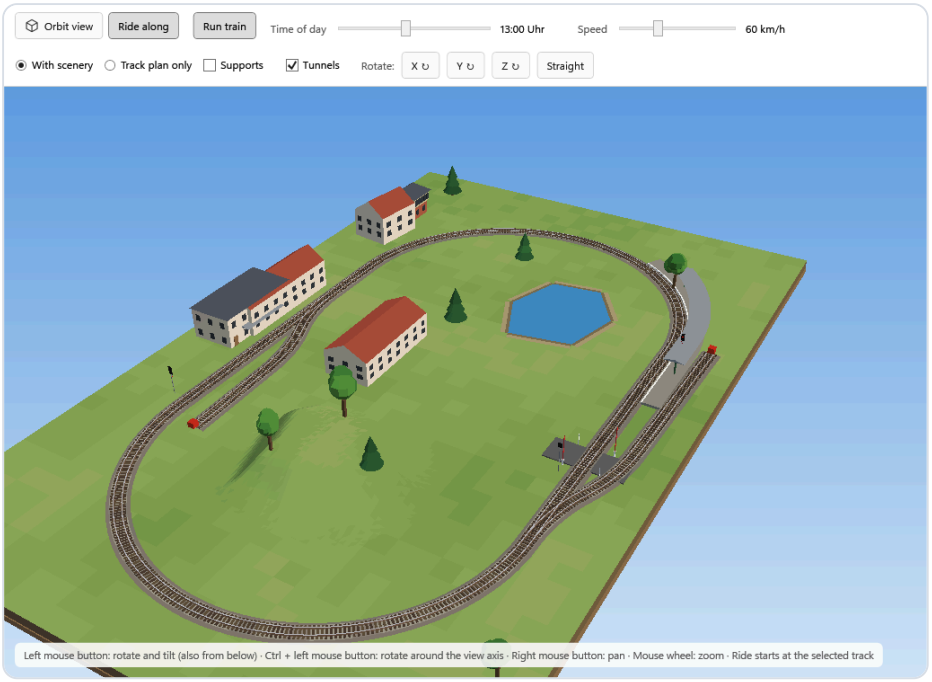


Figure 21 – Basic models in the 3D view.

The model editor PRO

Scenery items › New model opens the editor with an empty house. **From template…** uses a basic model or one of your models as a starting point. For a model in the plan, **Edit model** in the properties leads straight into the editor; a basic model is copied, the original stays unchanged.

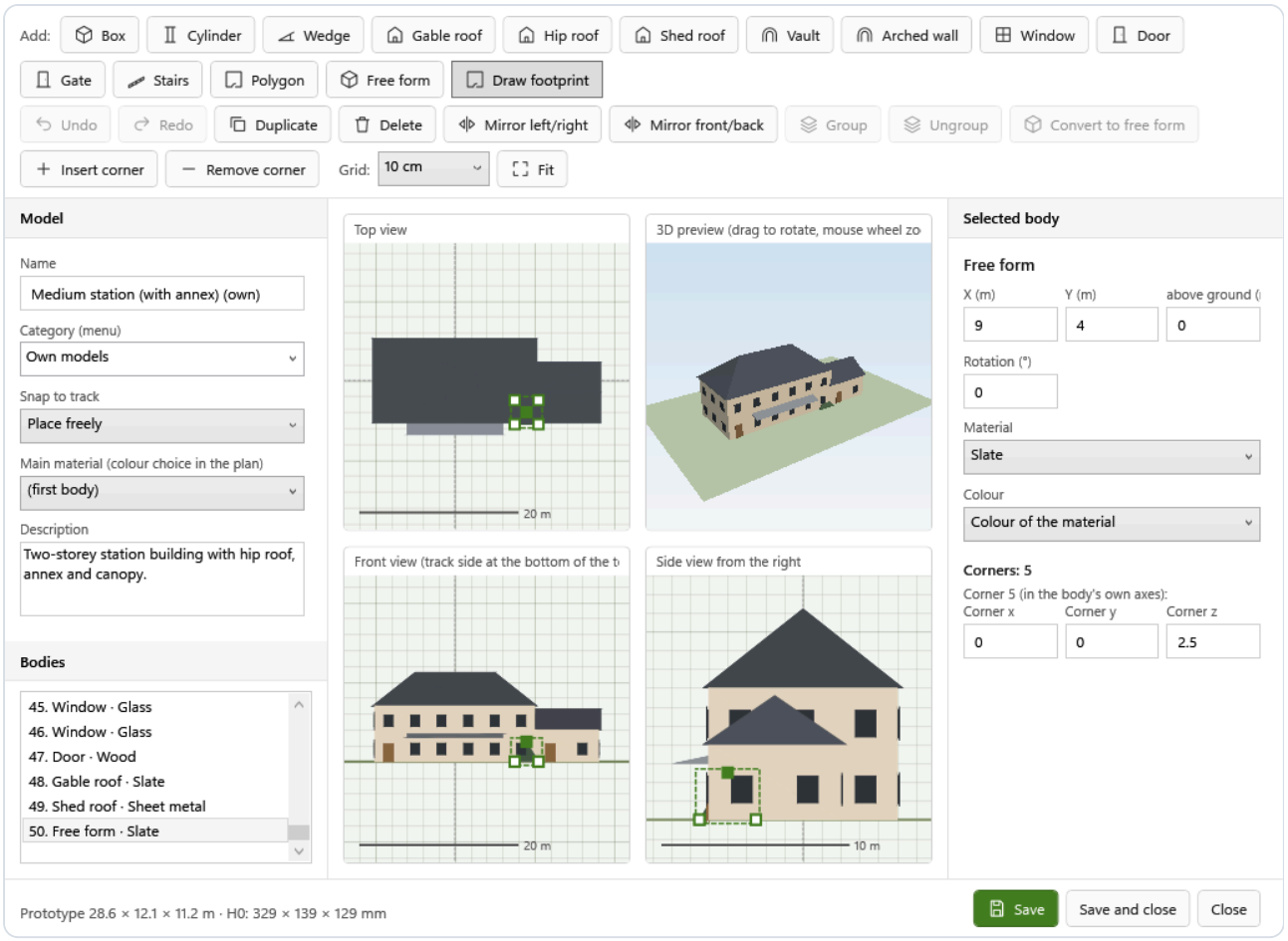


Figure 22 – The model editor: top view, 3D preview, front and side views; the model on the left, the selected body on the right.

A model consists of **bodies** – like building blocks. All dimensions are metres on the prototype; the program does the scaling.

Body	Used for
Box, cylinder, wedge	Main building, towers, chimneys, ramps. With **Top** a box becomes a pyramid, a cylinder a cone.
Gable, hip and lean-to roof	Placed on top of the selected body when added, with roof overhang.
Vault, arched wall	Tunnel tubes, bridge arches, portals; width and height of the opening are adjustable.
Window, door, gate	Placed on the front of the selected body.
Stairs	Steps of suitable height.

Polygon	Free floor plan with any number of corners; each corner can have its own height.
Free form	Every corner can be dragged in all three directions – for anything the other shapes cannot do.

Working in the editor

- **Select and move:** clicking a body selects it, with Ctrl or Shift several. Dragging moves on the set grid, the arrow keys by one grid step. The mouse wheel zooms, the right mouse button pans the view, **Fit** shows everything.
- **Exact values** are entered on the right: position, length, depth, height, rotation, material and colour.
- **Drawing a floor plan:** switch it on and click corner by corner in the top view; double-click or Enter closes the floor plan.
- **Editing corners:** for polygons and free forms the corners appear as small squares you can drag. **Insert corner** and **Remove corner** change their number.
- **Duplicate**, **Mirror**, **Group**, **Undo** and **Redo** as in the track plan.
- On the left you set **name**, **category**, **snapping to track** and the **main material**. At the bottom is the size on the prototype and at your gauge's scale.

> **Tip**
> The front view shows the side that is at the bottom in the top view. Build platforms and stations so that the track runs along that side.

Your own library, import and export PRO

Saved models are under **Scenery items › Own models** and can be inserted like basic models. On Windows they are files (`*.tmmodell`) in the folder `Documents\TrackMaster Pro\Modelle`, in the web version in the browser's storage.

Library... shows all your models: edit, edit as copy, insert, delete and export – individually as `*.tmmodell`, all together as ZIP or as a **3D file**: **STL** for the 3D printer and **OBJ** with colours for other 3D programs, each at your gauge's scale in millimetres. **Import...** reads model files and ZIP collections.

If a plan uses your own models, they are **saved with the plan**. Anyone who opens the plan sees them even without your library. **Add to library** brings such a model into your own library. The Android app inserts basic and plan models; the editor is not available there.

12 Room and background image

Will the layout actually fit in the room? Will the door still open, and can you reach every spot? On the **Room** tab you draw your room with doors, windows and furniture around the layout. A background image – such as a floor plan or a track plan from a book – helps with tracing.

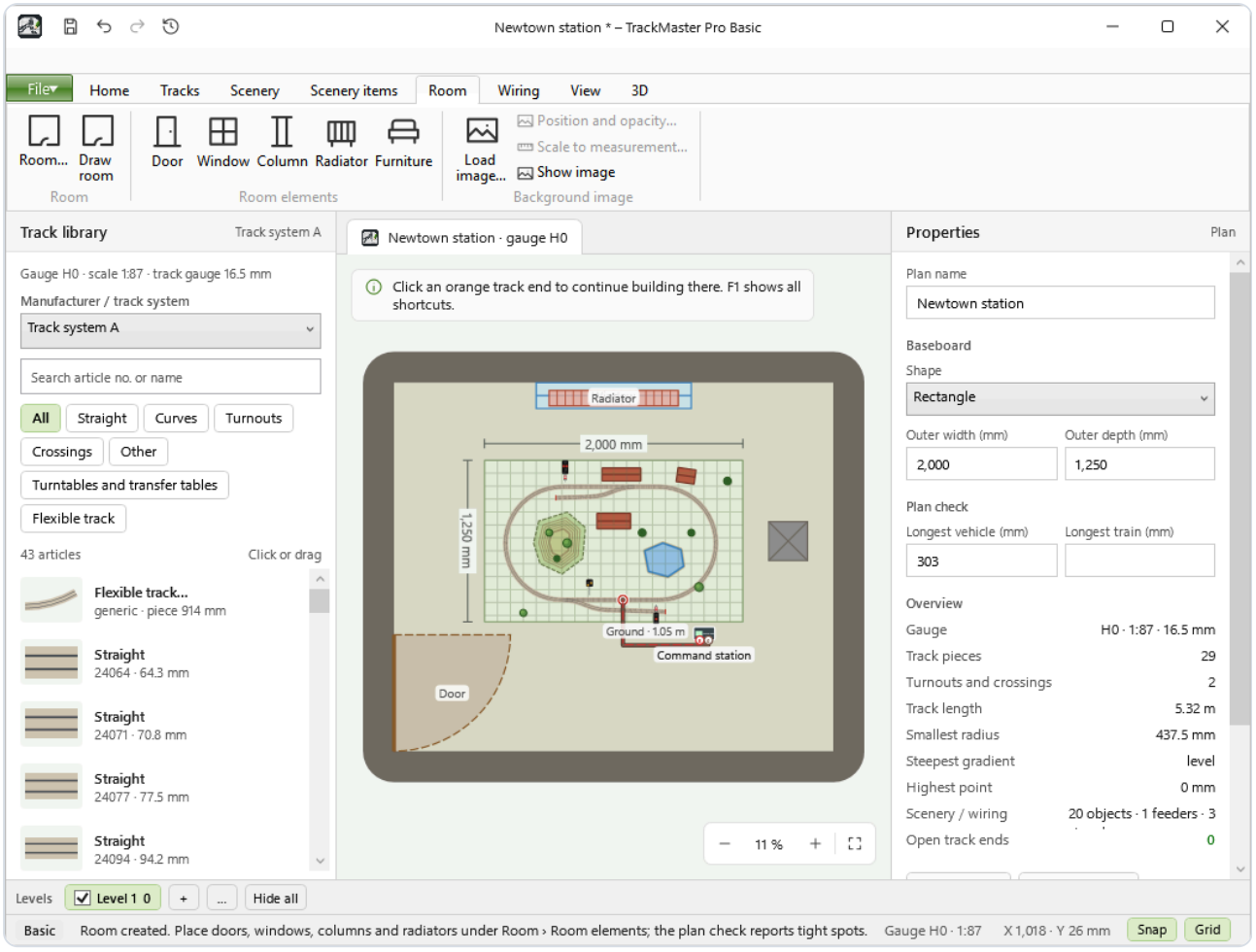


Figure 23 – The Room tab: room with door, window, radiator and column around the layout.

Creating a room

For a rectangular room, a window with a few numbers is enough:

1. Measure the room inside, from wall to wall, and decide where the layout will stand.
2. Click **Room › Room...**.
3. Enter the **inner width**, **inner depth** and **wall thickness** (all in mm).
4. Under **Inner corner X** and **Inner corner Y** state where the top left corner of the room is – measured from the top left corner of the baseboard.
5. Click **Create**. The walls appear in grey around the baseboard.

> **Example**
> The baseboard stands 50 cm from the left wall and 30 cm from the top wall. The room corner is therefore to the left of and above the baseboard: enter **−500** and **−300**. Minus means “left of” or “above” the baseboard.

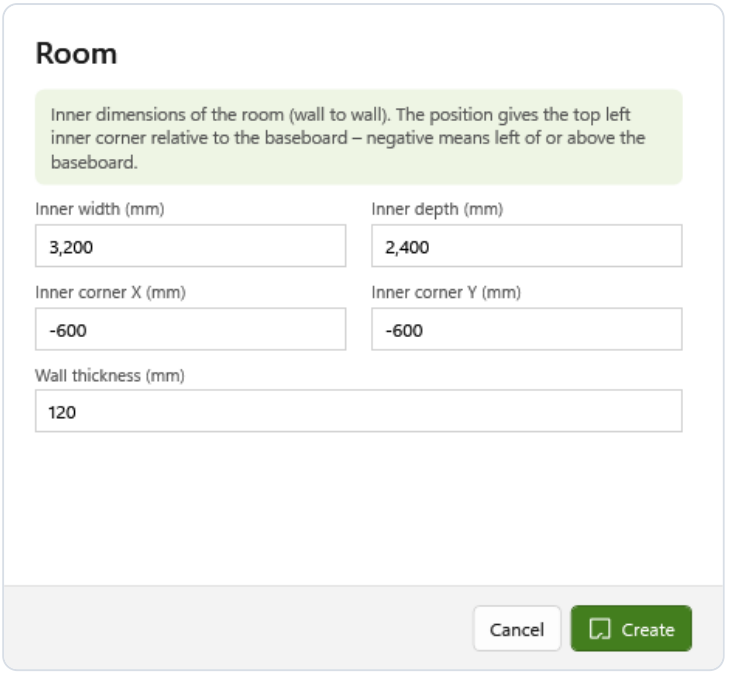


Figure 24 – Room window with inner dimensions, wall thickness and position.

If the room has alcoves or slanted walls, draw it with **Room › Draw room**: click all corners of the **inside of the wall** one after another. Double-click or Enter closes the room, Esc cancels. You select a room by clicking a wall; the properties show name, wall thickness and floor area in m².

Doors, windows and furniture

The **Room elements** group contains **Door**, **Window**, **Column**, **Radiator** and **Furniture**. Choose the element and click the spot in the plan, or draw a rectangle of the size you want. Then enter width, depth, height and rotation exactly in the properties. For a door, the plan shows the **swing area** – the space the door leaf needs when it opens. Esc ends the tool.

What the plan check reports in the room

Message	Meaning
Baseboard over the wall	**Error:** the baseboard sticks out through a wall – it will not fit in the room like this.
Door cannot open	**Error:** the swing area of a door lies under the baseboard.
Room element under the baseboard	**Warning:** a column, radiator or furniture stands under or against the baseboard. Windows may lie under the baseboard.

Aisle too narrow	**Note:** less than **60 cm** remain between baseboard and wall. That much space is comfortable for building and operating. If the baseboard stands directly against the wall (up to 15 cm away), this counts as intended.

Loading a background image

An image under the plan helps you rebuild a track plan from a book or plan the layout from a floor plan.

1. Click **Room › Load image…** and choose a photo or scan (PNG, JPEG, GIF or WebP).
2. The image now lies semi-transparent under the plan. Under **Position and opacity…** (web version: **Adjust image…**) you set where its top left corner is, how **wide** it is and how strong it appears (**opacity** in %). There you can also replace or remove it.
3. **Show image** hides and shows it. The image is saved with the plan.

Scaling the image to a measurement

A scanned image has some arbitrary size at first. To make it exact, measure a distance on it whose real length you know – such as a room wall:

1. Switch on **Home › Dimension** and click the start and end of the known distance in the image.
2. Click the new dimension line so that it is selected.
3. Click **Room › Scale to measurement…**. The window shows how long the line is now. Enter the **real length** in mm and click **Scale**.

The image is enlarged or reduced to match. You can then delete the dimension line and lay your tracks exactly on the image.

13 Wiring, signals and digital

A locomotive only runs when there is power in the rails. In this chapter you plan where the power comes from, where it gets into the track, where the rails are electrically separated – and let the program check that everything fits together. Signals, digital addresses and feedback sensors follow. You need no previous knowledge; the basics are in the first section.

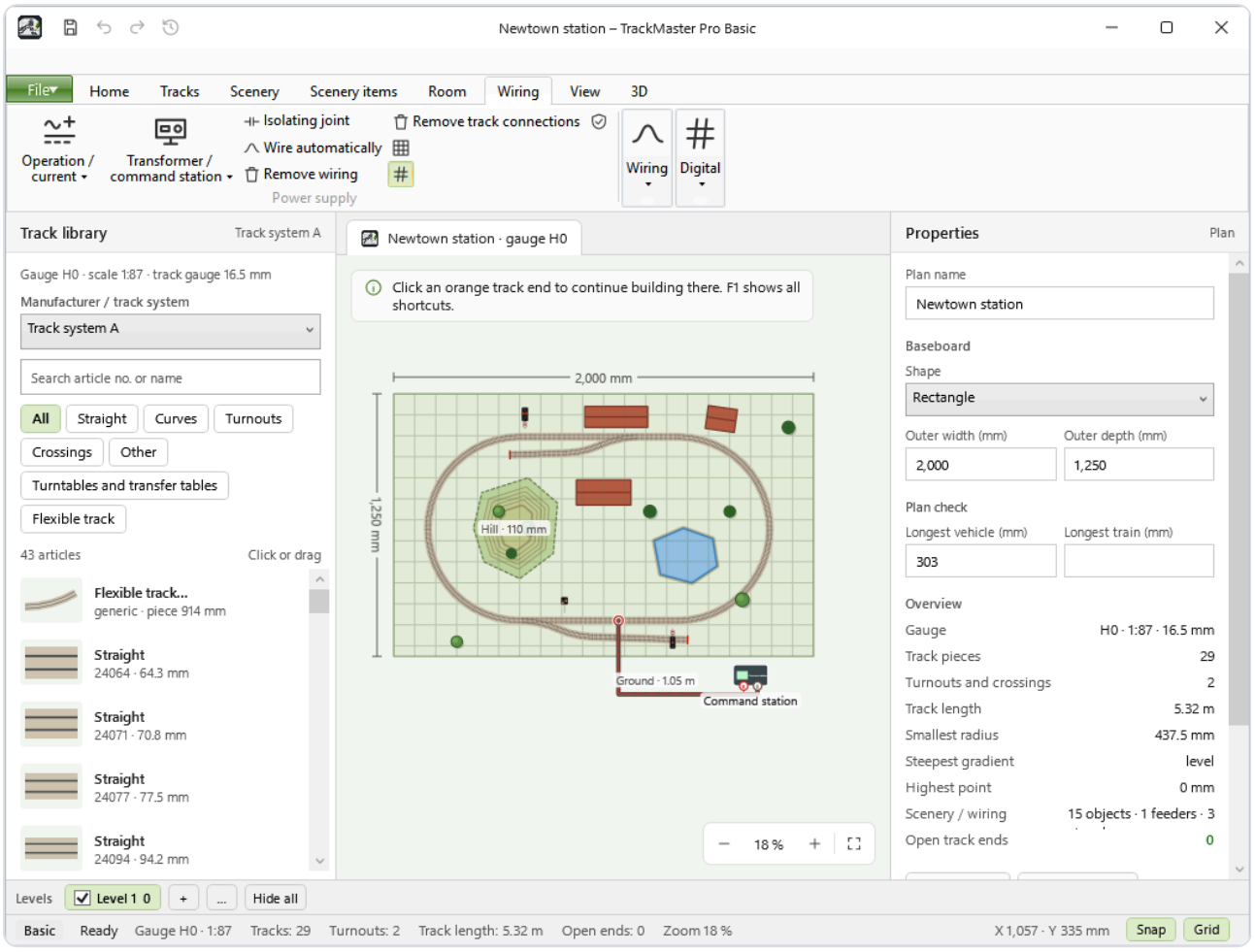


Figure 25 – The Wiring tab.

The basics in five minutes

Electricity only flows in a **circuit**: from the power source through one wire to the locomotive and back through a second wire. On a model railway the rails themselves are the wires. The locomotive picks up the power through its wheels (or a pick-up shoe).

- **Track power (B)** is the current that flows to the locomotive. In the program and on many devices the terminal is called “B”; the wire is usually **red**.
- **Ground (0)** is the return path to the power source. The terminal is called “0”; the wire is **brown** (centre rail) or **blue** (two-rail). Without ground no current flows – and nothing runs.
- **Feeder:** the spot where a wire is connected to the track. This is where power gets into the track.

- **Isolating joint:** a spot where the rails are electrically interrupted (with insulated rail joiners instead of metal ones). This creates separate **sections** that can be supplied or switched off individually.
- **Short circuit:** track power and ground touch directly, without a locomotive in between. The power source then shuts down. You want to avoid this – the program warns you.

Two-rail or centre rail?

There are two kinds of model railway track. Which one you have follows from the track system you chose.

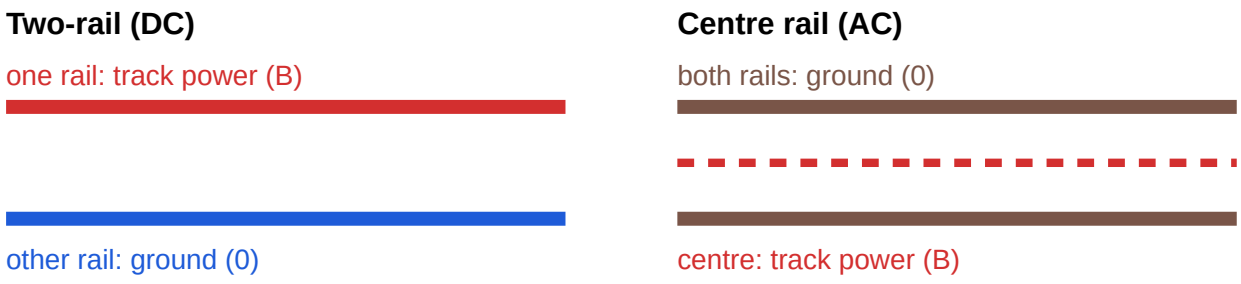


Figure 26 – Where track power and ground are.

	Two-rail (DC)	Centre rail (AC)
Track power	on one rail	on the small studs in the middle of the track
Ground	on the other rail	on both rails
Wire colours	red and blue	red and brown
What to watch	Within a section, track power must always be on the **same** rail. A **reversing loop** needs special wiring.	Polarity and reversing loops do not matter.

Analog or digital?

- **Analog:** a **power pack** (throttle) puts power into the track. The more power, the faster the locomotive runs. All locomotives in the same section run at the same time.
- **Digital:** a **command station** keeps power in the track all the time and sends commands with it. Each locomotive has its own address and obeys only its own commands. Large layouts add **boosters** – amplifiers that supply further sections with power.
- An **accessory transformer** supplies lights, turnouts and signals. It does not belong on the track.

Step by step: power to the track

This is how to supply a simple layout with power. The example is an oval with a command station.

1. Set operation and current

Click **Wiring › Operation / current**. Choose **Digital** or **Analog** and **DC** (two-rail) or **AC** (centre rail). If in doubt, leave the current at *Automatic (from track)* – it then follows your track system. The check, the polarity and the suggested wire colours depend on it.

2. Place the command station or transformer

Click **Wiring › Transformer / command station** and choose what to place – in digital operation a **command station** or **booster**, in analog operation a **power pack**, and in both an **accessory transformer**. Then click in the plan. The device appears as a box with two terminals: **B** (track power) and **0** (ground). You may also put it next to the baseboard, for example where it will really stand.

3. Lay the track power wire

1. Click the command station (or transformer).
2. Click it with the **right** mouse button and choose **Lay wire from here (track power)**. The same button is also in the properties on the right.
3. The wire starts at terminal B. Click points in the plan along which the wire is to run – for example along the underside of the baseboard.
4. Finally click **on a track** and finish with a **double-click** (in the web version also with **Done**).

Where the wire ends on the track, a **feeder** is created. It is shown in the plan as a small symbol at the track.

4. Lay the ground wire

1. Right-click the command station and choose **Lay ground wire from here (0)**.
2. Click points and let the wire end on a track – with two-rail track on the **other** rail from the track power. A double-click finishes.

A **ground connection** is created there. With centre rail it does not matter which rail; with two-rail the program makes sure track power and ground are on different rails.

> **Example: How many feeders?**
> Power loses strength over long distances because rail joiners do not conduct perfectly. As a rule of thumb there should be a feeder about every 2 to 3 metres. An oval with 6 m of track therefore gets two or three track power wires. Simply choose **Lay wire from here** several times; a track may also have several feeders. The check gives a note if there are too few.

Changing or deleting a feeder

- Click a feeder. The properties show which source it is connected to.
- **On the right rail (in track direction)** moves it to the other rail. You need this with two-rail track when the check reports “wrong rail”.
- Del deletes the feeder. If the wire is deleted, its feeder disappears too.

Isolating joints, circuits and reversing loops

Placing an isolating joint

Switch on **Wiring › Isolating joint** and click the spot where two tracks meet. An isolating joint is created there; a second click removes it again. Alternatively: right-click a track and choose **Isolations** (Windows: **Isolators and feedback sensors**) – there you switch the isolation on at each end individually. With two-rail track an isolating joint means “both rails insulated”, with centre rail “centre rail separated”.

What for? For example for a siding where a locomotive is to stand without power, for signals that switch off the power in front of them, for the boundary between command station and booster, or for a reversing loop.

Showing circuits

Wiring › Show circuits colours each section in its own colour. You see at once which tracks are electrically connected. **Label electrics only** hides heights and article numbers and writes the section numbers on the tracks instead.

Reversing loop

A **reversing loop** is a track arrangement in which a train turns round and comes back in the opposite direction – for example a loop at the end of a line. With **two-rail** track the left rail meets the right one: track power meets ground, a short circuit. The solution: put isolating joints at both ends of the loop and supply it via a **reversing module** (digital) or a **polarity reversing switch** (analog). The program detects reversing loops itself and reports them. With centre rail this problem does not exist.

Wire automatically

The quickest way to finished wiring: build the track plan, set the isolating joints, then **Electrics › Wire automatically** (also in the context menu). The program adds feeds with **track power and ground** to every section – several on long sections, spaced by the feeder rule – and runs the wires to the source.

- **Track system:** The program detects the track used (the one with the most track length). Centre rail (e.g. Märklin C and K track): track power to the centre rail, ground to the rails, no polarity; detectors via an isolated outer rail on the feedback module. Two-rail: track power and ground each to one rail, always the same side within a section; reversing loops get a reversing module, detectors run via the occupancy detector. If the selected current type does not match the track, it is set to “Automatic (from track)”.
- **Ground:** every feed gets its own ground wire to the track.
- **Layout on one level**, however many isolating joints: the **command station** feeds all sections.
- **Several levels:** one **booster** per level; the command station is only used for switching. Isolating joints alone never add boosters. For a level to get its own booster it must be a separate section, divided off by isolating joints.
- **DC (two-rail)**: every reversing loop is detected, missing isolating joints at its ends are set and the loop is fed via a **reversing module KS1, KS2 ...**. The module gets track power (B) and ground (0) at the top from the command station or booster.

- **Detectors are placed automatically:** in digital mode every section between isolating joints becomes a detection section with one detector (Melder 1, 2 …, consecutive contact); reversing loops excepted. You no longer need to place feedback sensors by hand. Detectors you placed yourself stay; running it again only replaces the automatic ones.
- Sections with a **detector** (digital) run via the **occupancy detector**; it hangs on the command station or a booster like a booster does. You can also place the occupancy detector yourself via **Electrics › Occupancy detector**.

Running it again replaces the automatically added wires, boosters and modules; wires you laid yourself stay. Everything can be undone in one step. **Electrics › Remove wiring** (also in the context menu) takes out everything automatic: wires, track connections, command station and boosters, reversing modules and occupancy detectors, detectors and the isolating joints set for reversing loops. Anything you placed yourself stays. **Electrics › Remove track connections** on the other hand only takes out all connections to the track – track power and ground, automatic or laid by hand – together with their wires; command station, boosters, modules, detectors and isolating joints stay.

Checking the wiring

Wiring › Check wiring examines the whole plan and lists everything unusual. Clicking a message shows the spot in the plan. **Errors** must be fixed, otherwise the layout will not run or there will be a short circuit. **Warnings** should be looked at. **Notes** are advice.

Message (in short)	What to do
Section … has no feeder	**Error.** This part of the track gets no power. Lay a track power wire from command station, booster or transformer to a track in this section.
Section … has track power but no return	**Error** (two-rail). The second rail is not connected to ground. Lay a ground wire to the other rail.
No ground on the track yet	**Warning** (centre rail). Lay a ground wire to one rail.
Feeder on the wrong rail	**Error** (two-rail). Track power and ground are on the same rail – short circuit. Click the feeder and switch **On the right rail**, or lay the wire to the other rail.
Track power (B) and ground (0) connected directly by wires	**Error.** A ground wire ends at the same spot as a track power wire. Let the ground wire end at its own spot on the other rail.
Ground (0) not connected	**Warning.** No wire is connected to terminal 0 of the source. Use “Lay ground wire from here”.
… feeder(s) – about one every … m is recommended	**Note.** The section is long; another feeder stops locomotives slowing down.
Section … is fed from several circuits	**Error.** Command station and booster (or two transformers) feed the same section. Put an isolating joint between them.
Reversing loop over … tracks	**Error** (two-rail). Put isolating joints at both ends and supply it via a reversing module (see above).

Ground of booster and command station not connected	**Warning.** Command station and booster need a common ground, for example via a brown distributor.
Accessory transformer with feeder	**Error.** An accessory transformer is meant for lights and turnouts. Traction power comes from command station, booster or power pack.
Power pack in digital operation / command station in analog operation	**Warning.** Device and operation do not match. Change the operation or choose another device.
Current does not match the track	For example AC set, but two-rail track laid. Change it under **Operation / current**.
Address … is used more than once	**Error.** Two turnouts or signals respond to the same digital address. Change one under **Addresses**.
Wire not connected at either end	**Note.** A wire ends in mid-air. Extend or delete it.

Laying wires and distributors

Besides the wires for traction power you can plan any other wire – for turnouts, signals, lights and feedback sensors. The program measures the lengths and turns them into a shopping list.

1. Click **Wiring › Lay wire** and choose what the wire is for. The suggested colour depends on it.
2. Click points in the plan. Start and end snap to command station and transformer (terminals B and 0), distributors, feedback sensors, signals and point motors.
3. A double-click finishes the wire (web version: also **Done**). Esc switches the tool off.

Use	Colour	Cross-section (suggestion)
Traction power	red	0.25 mm², 0.5 mm² for large gauges
Return / ground	blue (two-rail), brown (centre rail)	as traction power
Turnouts and signals	yellow	0.14 mm²
Lights and accessories	grey	0.14 mm²
Feedback sensors	violet	0.14 mm²
Other	black	0.14 mm²

Colour, cross-section and name of a wire can be changed in the properties. **Show wires** hides and shows all wires.

Distributors (**Wiring › Distributor…**) are terminal strips where many wires come together – in the program in **red, black, blue, yellow or brown**. All wires at a distributor are connected to each other. A typical example is a brown distributor where the ground of command station, boosters and track comes together.

Signals

Signals tell the driver whether the train may proceed. On a model railway they are mostly decoration, but they can also switch off the power in front of them.

1. Click **Wiring › Place signal** and choose the **type** and the **signal system** (country and design).
2. Click a track end. The signal is placed there.
3. Clicking an existing signal with the same tool removes it. Change type, system, name and address in the properties.

Type	Meaning
Main signal	Shows “stop” or “proceed” for the following section of line.
Distant signal	Stands at braking distance before the main signal and announces what it shows.
Exit signal	Main signal at the end of a station track; can also permit shunting moves.
Shunting signal	Small ground signal for shunting moves.
Block signal	Simple main signal on the open line that divides it into sections.

Signal systems: Germany (H/V colour-light, semaphore, Ks), East Germany (Hl), Austria, Switzerland (system L), France, the Netherlands, Great Britain and the USA. The signal is shown in the plan and in the 3D view with the matching signal head, each in its stop aspect.

Digital addresses

In digital operation every turnout and every signal gets an **address** – a number by which the command station speaks to it, like a house number. **Wiring › Addresses** shows all turnouts and signals in a list.

- Enter addresses one by one, or have them assigned in sequence with **Assign from**, starting from a number of your choice. With **only missing**, existing addresses are kept.
- Turnouts use one address, double slips and main signals two.
- Duplicate addresses are marked in red. You can also give each entry a name, for example “T3 station entry”.

Feedback sensors

Feedback sensors tell a digital control system (or control software on a computer) where a train is. Choose **Wiring › Detector** and the type:

- **Occupancy detector** – notices from the current drawn by vehicles that a track section is occupied. With two-rail track one rail is separated for this.
- **Contact track** – reports when a train passes a particular spot.
- **Reed contact** – a small switch between the rails, triggered by a magnet under the vehicle.

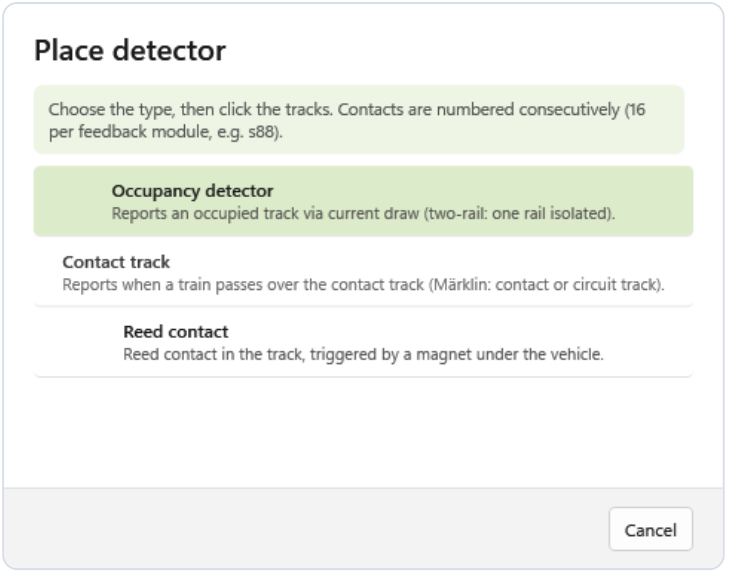


Figure 27 – Choosing the type of feedback sensor.

Then click the tracks. The contacts are numbered in sequence, 16 per feedback module. Clicking an existing sensor removes it. Change name, type and contact number in the properties. The sensors are numbered “Melder 1”, “Melder 2” and so on.

In digital operation an occupancy detector sits between the command station or booster and the track: the traction power flows through the detector into the track. In the properties, choose under **Traction power from (command station / booster)** where the detector gets its power from. A new detector automatically gets the command station or booster that already feeds its section, otherwise the first booster or the command station in the plan. The section then counts as fed – **Check electrics** no longer reports a missing feeder there. A red dashed line in the plan shows the connection, and the wiring plan lists “Traction power through Melder 3”.

Electrics › Wire detectors (also in the context menu: **Connect detectors to module**) connects every feedback sensor that has a contact number to the matching terminal of its feedback module with a violet wire. Contact 5 goes to input 5 of the first module, contact 20 to input 4 of the second. If there is no module for a contact yet, the program places a generic **occupancy detector** below the track plan (“Besetztmelder 1”, “Besetztmelder 2” …, one per 16 contacts). Track power (B, red) and ground (0, brown) are at the top of the occupancy detector: from there one wire each runs to the command station or booster you chose for the sensor under “Traction power from”. Feed and ground of the section are then fine. While the function is on, the wires follow every change: a new sensor, another contact number, a moved track or a deleted sensor. Clicking **Wire detectors** again (**Remove detector wires**) takes all these wires out again; the automatically added occupancy detectors go with them, feedback modules you placed yourself stay.

Numbering

Wiring › Number gives all feeders (G1, G2 …), feedback sensors (Melder 1, Melder 2 …) and wires (K1, K2 …) a number, from top left to bottom right. Write the same numbers on the real wires when building – then you will find every wire again later.

Wiring plan

Wiring › Wiring plan sums up all the wiring as a list – to print and take under the layout:

- power supply: every command station, booster and transformer with the number of its feeders and whether ground is connected,
- all sections with length and their feeders including the rail (left/right),
- all isolating joints, all feedback sensors with module and input,
- all wires with use, colour, cross-section, length, start and end,
- a **shopping list** of wire per colour and cross-section, with 15 % extra,
- the reversing loops.

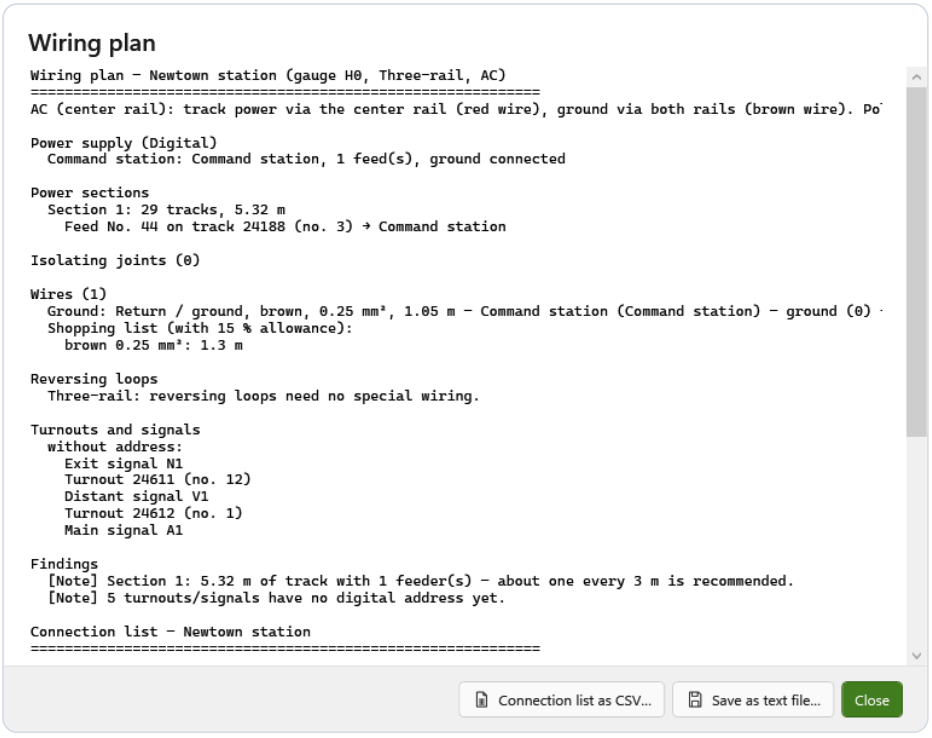


Figure 28 – Wiring plan.

The wiring plan can be saved as a text file in the Pro edition.

Routes

A **route** is a path from one signal to the next, with all turnouts in between in the right position. **Wiring › Routes** finds all such paths – up to the next signal in the direction of travel, a buffer stop or an open track end. Each route shows the turnouts with their position (straight, diverging, left, right), the length and the addresses. Routes that need the same turnout in different positions are marked as a conflict. This function is part of the Basic edition.

Export for control software PRO

If you control your layout with software on a computer, you would have to enter all turnouts, signals and addresses there again. **Wiring › Control export** saves you this: it shows an overview and saves turnouts, signals, addresses, feedback sensors, power sections and routes as **text**, **CSV** or **JSON**.

14 Plan check

Before you saw wood and buy track, the plan should be right. The plan check looks for places where something will go wrong later: tracks that touch, ramps that are too steep, curves that are too tight or a door that no longer opens.

Click **Home › Collision check** (Windows) or **Home › Plan check** (web version), or press F7. The result is a list of **errors**, **warnings** and **notes**. Clicking an entry selects the affected tracks and shows the spot.

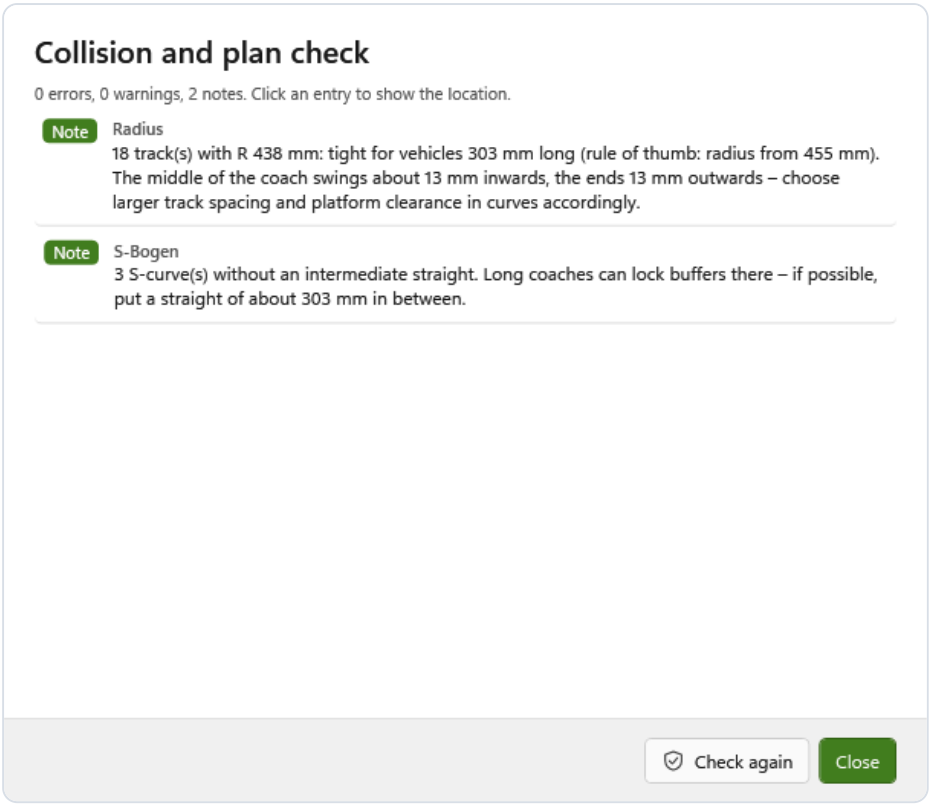


Figure 29 – Result of the plan check.

Check	What is reported
Track spacing and overlap	Tracks that touch or lie too close together. For tracks at different heights, what counts is whether a train fits underneath.
Baseboard edge	Tracks that stick out over the edge of the baseboard or into a cut-out
Gradient	Warning from 3 %, error from 5 %
Gradient in a curve	Warning if a curve rises more steeply than 2 % – in a curve a train pulls noticeably harder
Scenery	Tracks in buildings, in water or under a mountain without a tunnel
Open track ends	Ends without a connection and without a buffer stop – a train would fall off here

Module boundary	Tracks that run across the boundary of two modules in a modular layout
Radius for vehicles	Curves too tight for your longest vehicle
S-curve	Two tight curves in opposite directions without a straight piece between them; long coaches can lock buffers here
Room and aisle	Baseboard over the wall, door blocked, furniture under the baseboard, aisle under 60 cm – see chapter 12
Usable length	Sidings shorter than your longest train

For the vehicle checks, enter the **longest vehicle** in mm in the **plan properties** (click an empty spot in the plan; section **Plan check**). Without an entry the program uses a long passenger coach. If you enter the length of your longest train under **Longest train**, the program also checks whether it fits on every siding.

> **Error, warning, note**
> An **error** must be fixed, otherwise the layout cannot be built or run like this. A **warning** should be looked at. A **note** is advice – sometimes the spot is exactly as intended. You check the wiring separately with **Wiring › Check wiring** (chapter 13).

15 3D view

In the 3D view you see your layout in three dimensions, like a finished model – and you can even ride along in the cab.

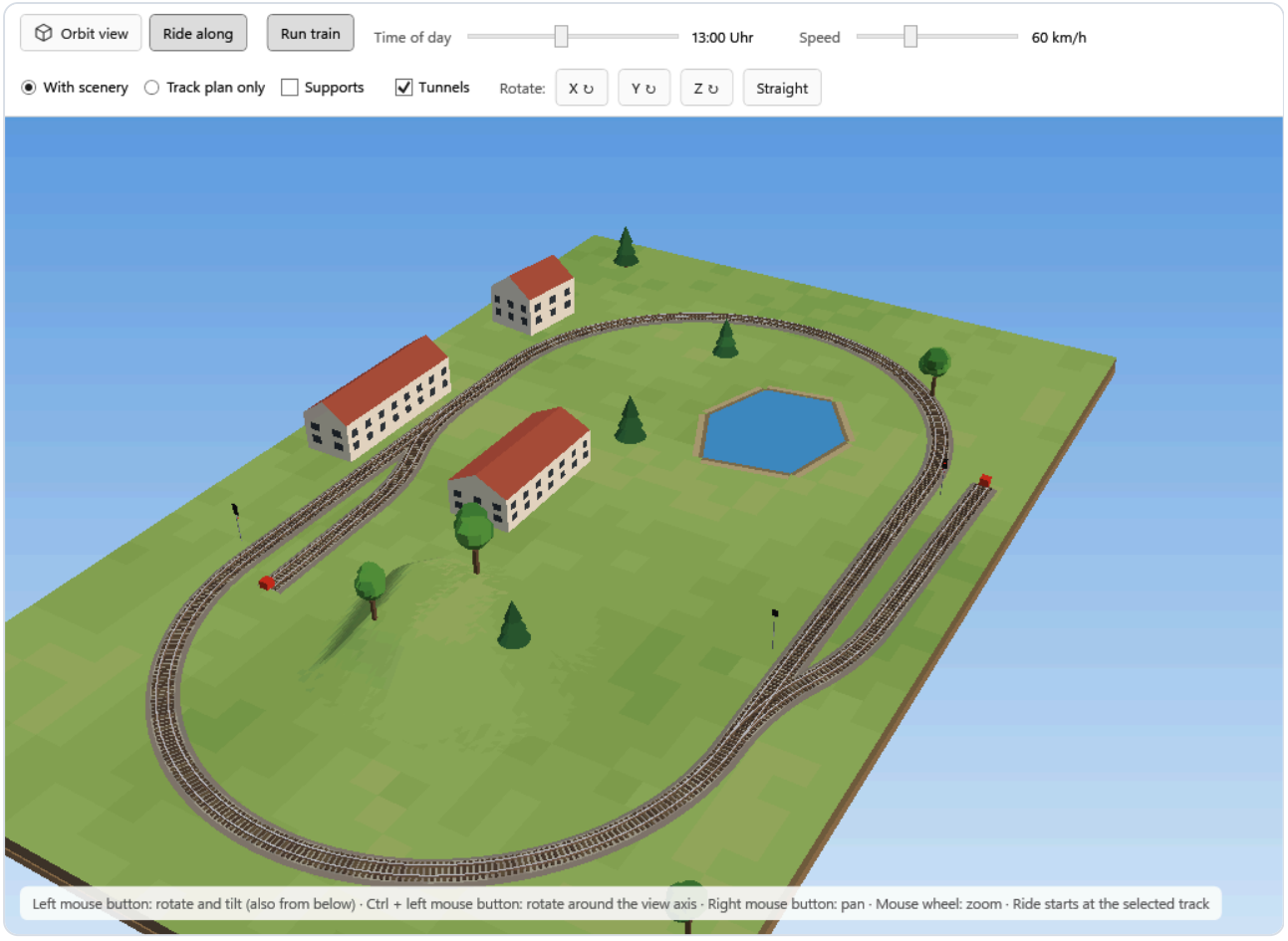


Figure 30 – The layout in 3D with terrain, buildings, trees and signals.

Looking around

Click **3D › 3D view**. Then:

- **Dragging with the left mouse button** turns and tilts the layout – even to below the baseboard.
- Ctrl + drag rotates about the viewing axis.
- The **right mouse button** pans the view, the **mouse wheel** zooms.
- The **X**, **Y** and **Z** buttons rotate in 90° steps. **Normal position** (web version: **Straight**) resets the view.

What is shown

- **With scenery** / **Track plan only**: show landscape, water, buildings and trees – or only baseboard and track.
- **Supports**: show or hide piers under raised track.

- **Tunnels**: when switched off, all tunnel sections disappear so you can see the rest of the layout clearly.
- Hidden levels are also missing in 3D.

Riding along

Select a track and click **3D › Ride along**. You travel over the track in the cab. Set the speed with **Speed**. Before every turnout the possible **directions** appear (◀ left, ▲ straight on, ▶ right); a click or the arrow keys ← ↑ → decide where to go at the next turnout. W switches the next turnout to its next position.

On Windows, **Run train** in the 3D window also sends a small train of a locomotive and three coaches round the layout, which you watch from outside.

Time of day

3D › Time of day sets the light from 5 am to 11 pm – with the position of the sun, evening sky and lit windows.

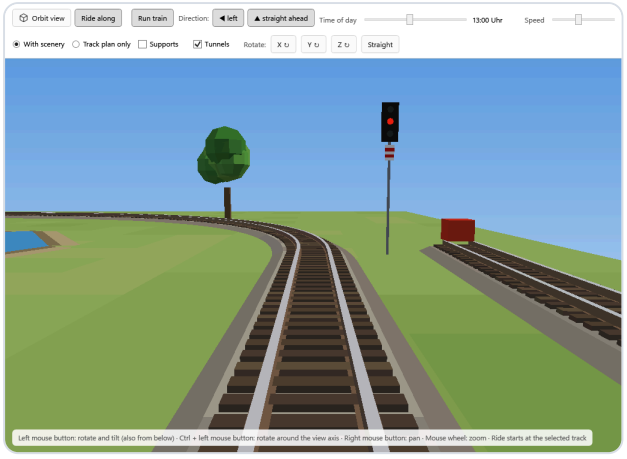


Figure 31 – Riding along in the cab.

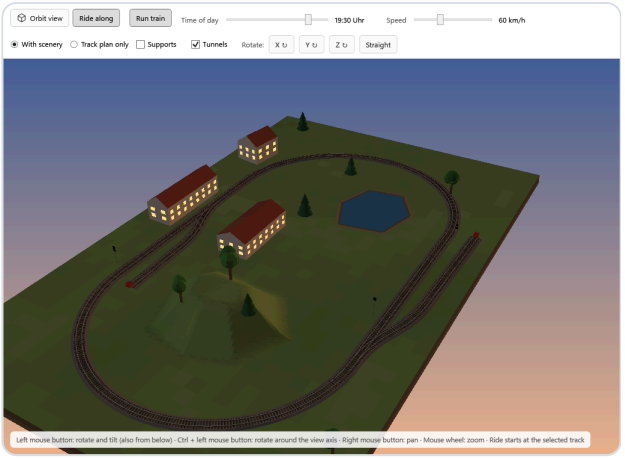


Figure 32 – Evening light.

> **Is the 3D view jerky?**
> On very large layouts **Track plan only** helps. If the picture stays black, update the graphics driver (Windows) or use a current browser (web version).

16 Saving, parts list, printing and export

Saving and opening

The Basic edition saves plans with up to **50 tracks**; the Pro edition saves larger plans. Every edition can **open** any plan.

	Windows	Web version
Saving	**File › Save** (Ctrl + S) and **Save as...**	**File › Save** (Ctrl + S) and **Save as...** (Ctrl + Shift + S)
File extension	`.trackmaster.json`	`.mpr` – in the browser's downloads folder or at the chosen location
Opening	**File › Open...** (Ctrl + O), **Recently opened**	**File › Open...** (Ctrl + O); opens `.mpr` and `.trackmaster.json`

The files can be exchanged between Windows, web and Android. On Windows and Android, choose “All files” in the file window to open a `.mpr` file. When a plan is opened, the track library switches to the plan's track system.

Automatic backup (Windows): while you work, the program regularly backs up the plan in the background. If it was not closed normally – for example after a power cut – it offers to restore the backed-up state at the next start. This does not replace saving.

Parts list PRO

Home › Parts list counts all tracks by article number, with description and quantity, and adds the accessories: point motors, signals, feeders, insulated joiners, feedback sensors and feedback modules. Flexible track is counted in whole lengths.

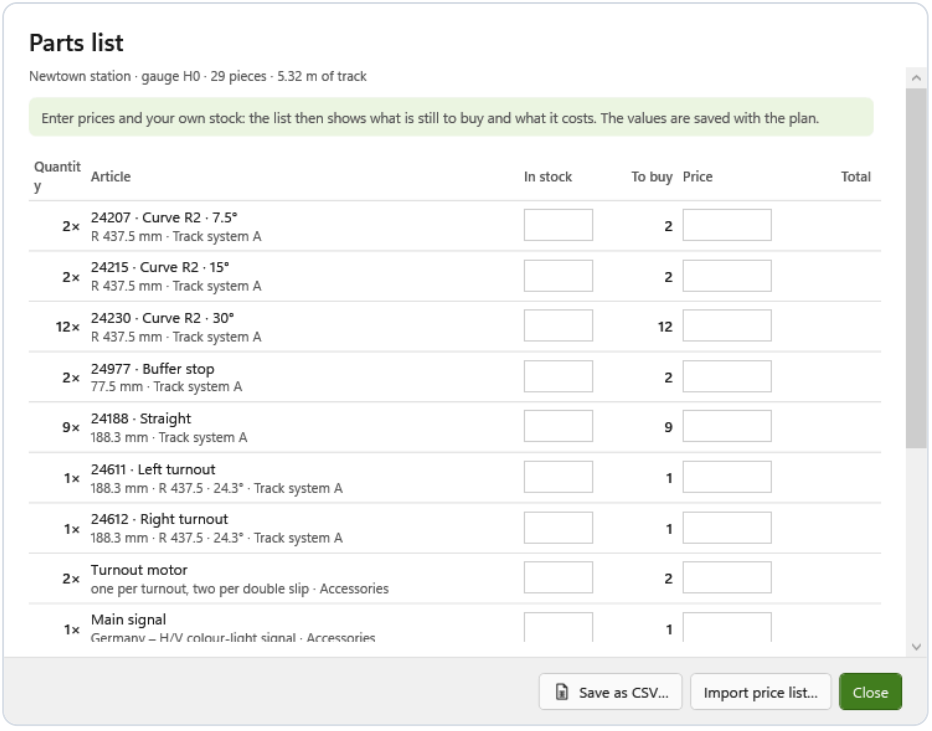


Figure 33 – Parts list with stock and prices.

What does the layout cost?

1. In the **In stock** column enter how many pieces you already have. The **To buy** column shows what is still missing.
2. In the **Price** column enter the unit price. The totals are shown at the bottom: **Still to buy**, **All bought new** and how many items still have **no price**.
3. Click **Close**. Prices and stock are saved with the plan.

Importing a price list: instead of typing in all prices, read in a text or CSV file with **Import price list...**, for example from a spreadsheet. Each line contains the **article number** (or the description) first and then the **price**, separated by semicolon, tab or comma:

```
10101; Straight 188 mm; 2.49
10202; Curve R2 30°; 2.79
Point motor; 19.90
```

With **Save as CSV...** (or **File › Parts list as CSV**) you take the list into a spreadsheet as a shopping list.

Printing PRO

File › Print... (Ctrl + P) offers two kinds:

- **Overview:** the whole plan reduced to one page (A4 landscape), with title and summary – to hang up or pass on.
- **Building template:** the plan at **full size** (1:1), spread over many pages. You glue the pages together, lay the template on the baseboard and transfer the track route with a pen or a pin.

How to get the building template right:

1. In the print window set the **actual size (100 %)** – not “fit to page”, otherwise the scale will be wrong.
2. On one sheet, measure the printed **100 mm test length** with a ruler. If it is correct, everything is right.
3. Each page has a **glue margin** (10 mm) that is covered by the neighbouring sheet, and **registration marks** every 100 mm. Lay the marks exactly on top of each other.
4. The page label (A1, A2, B1 …) tells you where a sheet belongs: letters for the columns, numbers for the rows.

If you want a PDF file instead of paper, choose “Save as PDF” or a PDF printer as the printer in the print window.

Export PRO

What	Where (Windows / web version)	What for
SVG graphic 1:1	**Export as SVG (1:1)…** / **Plan as SVG (1:1)**	Drawing at full size, for example for a plotter or cutting plotter
PNG image	**Export as PNG image…** / **Plan as PNG**	Image of the whole plan to show by e-mail or in a forum
DXF drawing	**Export as DXF (CAD)…** / **Plan as DXF (CAD)**	For CAD programs, CNC router and laser, 1:1 in mm, with separate layers (baseboard, track centre line, rails, buffer stop, scenery, room, text)
Parts list as CSV	**Parts list as CSV…** / **Parts list as CSV**	Shopping list for the spreadsheet

All of these are in the **File** menu or tab. In addition, the following can be saved from their windows: the **timber list** as CSV, the **wiring plan** as text, the **cutting templates** for helix, ribs and roadbeds as SVG files in a ZIP, the **ribs** as CSV and the **benchwork material list** as CSV (the benchwork is part of the Basic edition).

17 The Android app

The app offers almost all functions of the Windows program, adapted for finger operation. So you can plan on the tablet while out and about and carry on at the PC.

At the top is the ribbon with its tabs, at the bottom the track library as a strip with the level bar above it. Properties, plan check, helix, signals and the 3D view open as separate windows. Pro functions carry a padlock in the Basic edition.

Tab	Important commands
Room	Room, room elements and background image (from the gallery or a folder). Draw a room like terrain: tap the corners and finish with **Done**.
Wiring	Operation / current..., Transformer / command station..., Isolating joint, Place signal..., Detector..., Distributor..., Lay wire..., Addresses..., Show circuits, Label electrics only, Check wiring, Remove markers, Show wires, Number electrics, Wiring plan...
View	Fit, Grid, Track ends, Label, Heights, Snap, Levels..., rotate view, Design and font...
3D	3D view, Ride along, Evening light
File	New..., Open..., Save..., Sample layout, export as PNG and DXF, Print (PDF, also 1:1)..., Licence..., Appearance..., Sprache / Language, How to use...

Home, Tracks, Scenery and Scenery items contain the same commands as on Windows, including parallel track, height profile, lock, group and align.

Gesture	Effect
Tap a piece in the library	add it at the active track end
Tap	select; make a free track end active
Drag	pan the view; on a selected piece: move the piece
Two fingers apart / together	zoom in / out
Rotate two fingers	rotate the view; tapping the compass straightens it again
Long press	add a piece to the selection; on an empty area: selection frame
Double tap	properties; on an empty area: show the whole plan

The current plan is kept when you switch apps or lock the device. You save (Pro) and open files via the Android file picker – in device storage or in cloud storage.

18 Keyboard shortcuts and mouse commands

The program shows the list of all shortcuts with F1 (Windows: **Shortcuts**, web version: **How to use**).

Key	Function
Ctrl + N	New plan (Windows)
Ctrl + O	Open
Ctrl + S	Save (Basic up to 50 tracks)
Ctrl + P	Print PRO
Ctrl + F	Search track
Ctrl + Z / Ctrl + Y	Undo / Redo
Ctrl + A	Select all
Ctrl + D	Duplicate
Ctrl + C / Ctrl + V	Copy / Paste
Del	Delete
R / Shift + R	Rotate 15° right / left
F	Mirror
A	Switch connector
E	Next open track end
Enter	Add the last track again; while drawing: close the area
Arrow keys	Move the selection by 1 mm (Shift: 10 mm)
+ / − / 0	Zoom in / Zoom out / Whole layout
F7	Plan check
Esc	End tool, clear selection
F1	Shortcuts / How to use
W, ← ↑ → (3D)	Switch the turnout ahead of the train, choose direction

Mouse	Function
Click a track in the library	add it at the active track end
Drag a track from the library	place freely, snaps to nearby ends
Click an orange end	choose it as the active track end
Drag a track	move (with snapping)
Shift +click	add to the selection
Drag on an empty area	selection frame
Right mouse button	context menu; held down and dragged: pan the view
Middle mouse button, Space + drag	pan the view
Mouse wheel	zoom
Ctrl + mouse wheel	rotate the view
Click the compass	straighten the view
Double-click (room, terrain, wire)	finish drawing
Click a track of a group	select the whole group
3D: drag / Ctrl + drag	turn and tilt / rotate about the viewing axis

19 Questions and answers

A track does not snap into place.

Check that snapping is switched on: **Snap** at the bottom of the status bar (in the web version also under **View**). Tracks only snap to **free** ends; if the end is already used, choose the next open end with E.

An oval does not quite close.

Small gaps appear when straights of different lengths are mixed. Close the gap with **Tracks › Close gap** or draw flexible track (chapter 8).

I have changed too much.

Ctrl + Z takes back the last step. **History…** shows the last steps; a click jumps to an earlier state and forward again. On Windows the history is on the **Home** tab, in the web version it is an icon at the top of the title bar next to undo and redo.

Tracks have suddenly disappeared.

Their level is probably hidden. Tick it again in the level bar at the bottom, click **Show all** or choose **Show all** in the Levels window. If **Show scenery** is off, hills, water, buildings and trees are missing as well.

A track cannot be moved, rotated or deleted.

It is locked. Select it and click **Home › Lock**, or untick **locked** in the properties.

One click suddenly selects many tracks.

The tracks are grouped. **Home › Ungroup** separates them again.

“Section … has no feeder”.

No track power wire leads into this track section yet. Often there is an isolating joint in between that you forgot. Switch on **Wiring › Show circuits** to see the sections, and lay a wire from the command station to a track in this section with **Lay wire from here** (chapter 13).

“Ground missing” or “no return”.

The return path of the current is missing. Right-click the command station or transformer, choose **Lay ground wire from here (0)** and let the wire end on a rail – with two-rail track on the other rail from the track power.

“Feeder on the wrong rail”.

With two-rail track, track power is on the rail that already carries ground in this section. Click the feeder and switch **On the right rail (in track direction)**.

A note about the Pro edition appears when saving.

The Basic edition saves plans with up to 50 tracks. For larger plans install your licence under **File › Licence…** (chapter 3).

The licence is reported as invalid.

The licence file must not be changed – not even by opening and saving it in a text editor. If the licence is bound to a computer, it only applies there; if an expiry date is set, only up to that day. Contact the publisher with your computer ID.

The licence is valid but “not activated on this device”.

The licence is already running on two other devices, or the device has not been connected to the internet for 14 days. Release another device or connect to the internet and click **Activate now** in the licence window (section Online activation).

Windows reports “Windows protected your PC” during installation.

Click **More info** and then **Run anyway**. The message appears for programs that Windows has not seen often yet.

Android does not install the app.

Allow the app you use to open the file (for example “Files” or the browser) to install unknown apps. If an older version with a different signature was installed, uninstall it first.

The background image does not match my dimensions.

Draw a line over a known distance in the image with **Home › Dimension**, select it and call **Room › Scale to measurement…** (chapter 12).

The plan check says the door cannot open.

The door's swing area lies under the baseboard. Move or rotate the door, or make the baseboard smaller at that point.

How much does my layout cost?

Open the parts list (Pro), enter stock and prices or import a price list. The total is at the bottom (chapter 16).

The 3D view stays black or is jerky.

Update the graphics driver or the browser. On very large layouts **Track plan only** helps.

Where is my data?

Plans are wherever you saved them. On Windows the licence, settings and automatic backup are under *%APPDATA%\TrackMaster Pro*; the web version stores settings and your own models in the browser. The program does not send plans to third parties; at start-up it only retrieves the current version number.

20 Glossary

Term	Meaning
Accessory transformer	Transformer for lights, turnouts and signals; does not belong on the track.
Active track end	The orange-marked free end where the next track is added.
Address	Number by which a command station speaks to a turnout, signal or locomotive.
Analog operation	A power pack controls the voltage in the track; all locomotives in the section run at the same time.
Baseboard	The surface the layout is built on, usually a wooden board on a frame.
Booster	Amplifier in digital operation that supplies another section with power. An isolating joint belongs between booster and command station sections.
Buffer stop	End of a track where the train stops.
Centre rail	Kind of track with studs in the middle of the track for the track power; both rails are ground. Operated with AC.
Clearance height	Minimum height between two tracks on top of each other so that vehicles fit underneath; depends on the gauge.
Command station	Heart of digital operation: produces power and commands for locomotives, turnouts and signals.
Connector	One end of a track piece; a turnout has three, a crossing four.
CSV file	Simple table file that any spreadsheet can open.
Curve	Curved track piece with a fixed radius and angle, for example 30°. Twelve 30° curves make a circle.
Digital operation	A command station keeps power in the track and sends commands with it; each locomotive listens only to its address.
Distributor	Terminal strip where many wires are joined, for example the common ground.
DXF file	Drawing format for CAD programs, CNC routers and laser cutters.
Feedback sensor	Tells the digital control whether a track is occupied or a train passes a spot.
Feeder	Spot where a wire from command station, booster or transformer ends on the track and brings power into it. Created with “Lay wire from here”.
Fiddle yard	Hidden storage yard, usually under the layout, where trains wait.
Flexible track	Track in lengths that can be bent and cut as required.

Gauge	Size of the model railway, for example H0, N or Z; determines track gauge and scale.
Gradient	How steeply a track rises, in percent: 2 % means 2 cm of height per 1 m of line.
Ground (0)	Return path of the current to the power source; terminal “0”, wire brown (centre rail) or blue (two-rail).
Group	Combined tracks that are selected and moved together with one click.
Helix	Spiral rising line to climb large heights in little space.
Isolating joint	Insulated rail joint that separates two power sections.
Level	“Storey” of the layout with its own base height that can be shown and hidden.
Module	Rectangular layout part that can be combined with others to form a layout.
Open track end	End of a track piece without a connection; marked orange in the plan.
Properties	Area on the right of the window; shows and changes the values of the selected track or object.
Radius	Distance from the centre of the circle to the middle of a curved track; the smaller, the tighter.
Rail top	Reference height for all height values in the plan.
Reversing loop	Track arrangement in which a train turns and comes back in the opposite direction; with two-rail track it needs special wiring, otherwise there is a short circuit.
Rib	Vertical board on the frame on which the roadbed rests in open-frame construction.
Ribbon	Bar at the top of the window with all commands, arranged in tabs.
Roadbed	Board under the track that carries it at the desired height.
Route	Path from one signal to the next with all turnouts in the right position.
Scale	Ratio between model and prototype; in H0, 1 cm on the model is 87 cm in reality (1:87).
Section (power section)	Tracks that are electrically connected. Sections are separated from each other by isolating joints.
Short circuit	Track power and ground touch directly; the power source shuts down.
Side rail, cross member	Outer frame and cross bracing of the benchwork.
Siding	Track ending at a buffer stop where trains or wagons are parked.
Stop aspect	Signal indication meaning “stop”.

Tab	A tab on the ribbon (Home, Tracks, Scenery …); a click shows its commands.
Track power (B)	The current that flows to the locomotive; terminal “B”, wire usually red.
Track system	A manufacturer's track range; its pieces fit together exactly.
Turnout	Track piece where a track divides into two (or three) directions.
Two-rail	Kind of track in which the two rails carry track power and ground. Operated with DC.
Usable length	The maximum length of a train on a track.

TrackMaster Pro 1.5.0 · User Manual, November 2026. Subject to change. All dimensions without guarantee; check dimensions against the original before cutting and buying.